

Layers of immunity: Deconstructing the *Drosophila* effector response

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Faustine Ryckebusch, Yao Tian, Mylene Rapin, Fanny Schüpfer, Mark A Hanson , Bruno Lemaitre 

Global Health Institute, Ecole Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland • Centre for Ecology and Conservation, University of Exeter, Penryn, United Kingdom

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This work provides one of the first **important** attempts to look at *Drosophila* immune responses against bacterial, viral, and fungal pathogens in a way that combines the roles of four major arms in immunity (Imd signaling, Toll signaling, phagocytosis, and melanization) rather than studying them separately. The findings are **convincing**, and the tools provided can be used as they are, or built upon, in various contexts.

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Abstract

The host innate immune response relies on the cooperation of multiple defense modules. In insects and other arthropods, which have only innate immune mechanisms, four main immune-specific modules contribute to defense against microbial invaders: the Toll pathway, the Imd pathway, the melanization response, and phagocytosis by plasmatocytes. Our present understanding of their relative importance remains fragmented, as their contribution to host defense has never been simultaneously assessed across a large panel of pathogens. Here, we have taken advantage of newly-described immune mutants in a controlled genetic background to systematically delete these four immune modules individually, in pairs, or all four simultaneously. Surprisingly, flies simultaneously deficient in all four immune modules are viable, homozygous fertile, and display no overt morphological defects, suggesting these immune mechanisms are not strictly required for organismal development. With this new set of lines, we assessed the individual and collective contribution of each module to host defense against five viruses, three fungi, eight Gram-positive bacteria, and eight Gram-negative bacteria. Our findings show that these four modules largely function independently or additively in host defense, although synergistic effects can occur for select pairs of modules. Our study confirmed the importance of the Imd pathway against Gram-negative bacteria and the Toll pathway against Gram-positive bacteria and fungi, largely via the induction of effectors such as antimicrobial peptides (AMPs) and Bomanins (Bom), but also reveals an important role of melanization against viruses, and a contribution of phagocytosis against various germs. Additionally, by examining microbial load kinetics in different mutants, we provide insight into how these modules contribute to tolerance or resistance against specific microbes. Our study provides insights into the architecture of the *Drosophila* immune system, revealing differential requirements of immune modules according to each

pathogen. The set of immune deficient lines provided here offers tools to better assess the role of these immune modules in host defense.

Introduction

The immune system is made up of a network of tissues and circulating cells whose primary function is to prevent and limit pathogenic infections (Danilova, 2006 [↗](#); Pradeu et al., 2024 [↗](#); Schmid-Hempel, 2021 [↗](#)). Although immunity has been studied in many animal species, it has been extensively characterized in only a few, such as the nematode *Caenorhabditis elegans*, the fruit fly *Drosophila melanogaster*, zebrafish, rodents, and humans. In recent years, this traditional understanding of the immune system as a specialized group of cells and tissues has gradually changed to a more complex picture of highly redundant multilevel defense modules. The last few decades have revealed a great deal about the molecular mechanisms regulating each of these immune modules, from recognition and signaling to effector production (e.g. phagocytosis, antimicrobial peptides) (Pradeu et al., 2024 [↗](#)). However, an immune response encompasses the concomitant activation of several modules and there has been little investigation of how modules contribute collectively to host defense. Immune research has traditionally focused on modules in isolation due to technical limitations. This article aims to fill this gap by analyzing the specific contributions of the main immune modules to the collective *Drosophila* systemic response.

The *Drosophila* systemic immune response consists of a set of humoral and cellular reactions in the hemolymph (insect blood) that occur when flies are systemically infected by bacteria, viruses, or fungi (Liegeois and Ferrandon, 2022 [↗](#); Westlake et al., 2024 [↗](#)). A first facet of the systemic immune response is the production of host-defense effectors by the fat body and hemocytes that are secreted into the hemolymph. Among the best-characterized effectors are the antimicrobial peptides (AMPs), whose expression is rapidly induced to incredibly high levels upon infection in a race to control invading microbes (Hanson et al., 2019 [↗](#); Imler and Bulet, 2005 [↗](#)). AMPs are small cationic peptides that exhibit antimicrobial activity and directly participate in microbial killing. Beyond AMPs, many other proteins and host defense peptides are secreted into the hemolymph, creating a hostile environment for invading pathogens through a variety of mechanisms. The systemic antimicrobial response is predominantly regulated at the transcriptional level by the Toll and Imd pathways (De Gregorio et al., 2002 [↗](#); Lemaitre et al., 1996 [↗](#)). In *Drosophila*, the Toll pathway is activated by cell wall components (fungal glucans and Lysine-type peptidoglycan), as well as microbial proteases (Gottar et al., 2006 [↗](#); Issa et al., 2018 [↗](#); Leulier et al., 2003 [↗](#); Vaz et al., 2019 [↗](#)). Toll signaling culminates in the activation of two NF- κ B transcription factors, Dif and Dorsal, which regulate a large set of immune genes (Ip, 1993 [↗](#); Lemaitre et al., 1996 [↗](#)). These encode small peptides such as the antifungal peptide Drosomycin, the Toll-regulated Bomanin peptides, and many other proteins (e.g. serine proteases, serpins, lipases) (Clemmons et al., 2015 [↗](#); De Gregorio et al., 2002 [↗](#); Fehlbaum et al., 1994 [↗](#); Ligoxygakis et al., 2003 [↗](#)). Flies lacking the Toll pathway are viable but highly susceptible to infection by Gram-positive bacteria and fungi (Buchon et al., 2009 [↗](#); Lemaitre et al., 1996 [↗](#); Rutschmann et al., 2002 [↗](#)). The Imd pathway is activated by DAP-type peptidoglycan produced by Gram-negative bacteria and a subset of Gram-positive bacteria (e.g. *Bacillus*) (Kaneko et al., 2004 [↗](#); Lemaitre et al., 1995 [↗](#); Leulier et al., 2003 [↗](#)). The binding of peptidoglycan to receptors of the Peptidoglycan Recognition Protein family (PGRP-SD, PGRP-LC, PGRP-LE) initiates an intracellular signaling cascade, which ultimately activates the NF- κ B factor Relish (Choe et al., 2005 [↗](#); Iatsenko et al., 2016 [↗](#); Kaneko et al., 2006 [↗](#); Kaneko et al., 2006 [↗](#); Westlake et al., 2024 [↗](#)). The Imd pathway regulates the expression of many genes, including those encoding antibacterial peptides and serine proteases (De Gregorio et al., 2002 [↗](#); Lemaitre et al., 1997 [↗](#)). Imd deficient flies are viable but susceptible to Gram-negative bacterial infection.

In addition to humoral pathways, melanization and phagocytosis are two complementary mechanisms that contribute to host survival upon systemic infection. Melanization is an arthropod-specific immune response resulting in rapid deposition of the black pigment melanin at wound or infection sites and the concomitant production of microbicidal reactive oxygen species (Cerenius et al., 2008 [↗](#); Nappi et al., 2009 [↗](#); Westlake et al., 2024 [↗](#)). The melanization and Toll pathways are co-activated at the level of the serine proteases Haya and Persephone, which in turn activate Spatzle-processing enzyme (SPE) and serine proteases critical to cleavage of prophenoloxidases (PPOs) (Dudzić et al., 2019 [↗](#); Nakano et al., 2023 [↗](#); Shan et al., 2023 [↗](#)). This is collectively referred to as the Toll-PO SP cascade (Westlake et al., 2024 [↗](#)). The melanization reaction itself relies on the oxidation of phenols, resulting in the polymerization of melanin. Two PPOs (PPO1 and PPO2) are the primary catalysts for melanin synthesis during the melanization response to systemic infection. Flies deficient for both *PPO1* and *PPO2* are viable, but lack hemolymph phenoloxidase (PO) activity and exhibit susceptibility to certain Gram-positive bacteria and fungi (Binggeli et al., 2014 [↗](#); Dudzić et al., 2015 [↗](#)).

Phagocytosis by both sessile and circulating plasmatocytes (*Drosophila* macrophage-equivalent cells) is thought to provide a complementary and important host defense (Melcarne et al., 2019 [↗](#); Ulvila et al., 2011 [↗](#)). The use of hemocyte-deficient flies (via targeted expression of a pro-apoptotic gene in plasmatocytes) has shown that hemocytes contribute to survival upon systemic infection to certain bacterial species (Charroux and Royet, 2009 [↗](#); Defaye et al., 2009 [↗](#); Stephenson et al., 2022 [↗](#)). Phagocytosis of bacteria greatly relies on two transmembrane receptors, NimC1 and Eater (Bretscher et al., 2015 [↗](#); Kocks et al., 2005 [↗](#); Kurucz et al., 2007 [↗](#); Melcarne et al., 2019 [↗](#)). *NimC1*, *Eater* double mutant larvae are viable and their hemocytes can effectively encapsulate and melanize macroparasites; however, their hemocytes are non-sessile and nearly totally phagocytosis deficient. While *NimC1*, *Eater* larvae have elevated hemocyte numbers, adult flies have few hemocytes and fail to perform phagocytosis, providing a good tool to assess the role of cellular response (Melcarne et al., 2019 [↗](#); Melcarne, 2020 [↗](#)).

We have scattered circumstantial evidence that each of these four immune modules are critical for defense against certain pathogens. However, we lack a global view of how, when, and to what extent each of these four immune modules contribute to host defense against specific pathogens, as they have never been simultaneously assessed. Moreover, the extent to which individual findings can be generalized to broad classes of pathogens remains unclear.

In this article, we have used a specific set of mutations principally affecting phagocytosis (*NimC1*, *Eater*), melanization (*PPO1*, *PPO2*), the Toll pathway (*spz*), and the Imd pathway (*Rel*) in a controlled genetic background to address their individual roles in host defense. We then recombined these and other mutations to generate double and even quadruple mutants of the four immune modules. This set of tools allowed us to systematically compare the survival of flies deficient in these responses against a broad panel of viruses, fungi, Gram-positive bacteria, and Gram-negative bacteria, to reveal the relative contributions of each immune mechanism to resistance against these pathogens.

Materials and methods

Insect stocks

Precise details of *Drosophila* stocks used in this study are provided in **Table 1** [↗](#). To minimize the influence of genetic background as a variable in this study, mutations were isogenized into the *DrosDel iso w¹¹¹⁸* genetic background (BDSC #5905) as described by Ferreira et al. (Ferreira et al., 2014 [↗](#); Ryder et al., 2004 [↗](#)). Isogenized lines were generated and used for every mutant except the double mutant Toll and Imd (*Rel^{E20}*, *spz^{rm7}*) line, for which homozygous isogenized flies were not viable in our hands. Double mutants for *NimC1¹*; *Eater¹* were poorly viable when isogenized in

the *DrosDel iso w¹¹¹⁸* background, and so we used both isogenic and non-isogenic lines over the course of our study to facilitate investigation. Furthermore, *NimC1¹*; *spz^{rm7}*, *Eater¹* triple mutants were homozygous lethal regardless of isogenization, and so could not be included. In addition to the immune module mutants, the Δ *AMP14* line bears a knockout of 14 different AMP genes, which includes the AMP families Drosomycin, Metchnikowin, Cecropin, Defensin, Drosocin, Attacin, and Diptericin, as described previously (Carboni et al., 2022 [↗](#); Hanson et al., 2019 [↗](#)). The *Bom^{155C}* mutation was characterized in Clemmons et al. (Clemmons et al., 2015 [↗](#)) and isogenic flies were generated in Hanson et al. (Hanson et al., 2019 [↗](#)). For larval experiments, GFP-labelled CyO or TM3 balancers were used to enable selection of homozygous mutant larvae.

Microorganism cultures

Microbe strain information, microorganism classifications, and culture conditions are listed in **Table S1** [↗](#). Bacterial cultures and *Candida albicans* were grown overnight shaking in appropriate media at 200rpm. *Aspergillus fumigatus* fungus was grown on Malt Agar at room temperature until sporulation. The entomopathogenic fungus *Beauveria bassiana* strain R444 was provided by Andermatt as spore preparations (BB-PROTEC) which were directly used in natural infections or dissolved in PBS for septic injury. Viruses DCV and FHV were kindly provided by Carla Saleh (Pasteur Institute, Paris), produced in S2 cells. The supernatant of S2 cells was titrated before being used for fly infections. Viruses SINV, DXV and IIV-6 were kindly provided by Ronald van Rij (Radboud University Medical Center, Nijmegen).

Heat-killed microbes were prepared by two repeats of boiling at 95°C for 30 minutes then freezing at -20°C for 30 minutes, before storage long term at -20°C. Microbe preparations were streaked onto agar plates to verify full efficiency of heat-killing before use in experiments.

Gene expression

Flies were inoculated by pricking at the junction of thoracic pleura with a needle dipped in a mixed pellet containing a 1:1 mixture of OD₆₀₀ = 200 heat-killed *Escherichia coli* and *Micrococcus luteus* (final OD₆₀₀ = 100 for each), and frozen at -20°C 6h, 12h and 24h post-infection. Total RNA was then extracted from pooled samples of five flies using TRIzol reagent per manufacturer's protocol and resuspended in MilliQ dH₂O. Reverse transcription was performed using PrimeScript RT (Takara) with random hexamer and oligo dT primers. Quantitative PCR was performed on a LightCycler 480 (Roche) in 96-well plates using Applied Biosystems PowerUP SYBR green Master Mix (Applied Biosystems). Data points represent pooled samples from three replicate experiments. Error bars represent one standard deviation from the mean. Statistical analyses were performed using one-way ANOVA with Holms-Sidak multiple test correction. Primers used in this study were:

- Drs-F 5'- CGTGAGAACCTTTTCCAATATGAT-3'
- Drs-R 5'- TCCCAGGACCACCAGCAT-3'
- DptA-F: 5'-GCTGCGCAATCGCTTCTACT-3'
- DptA-R: 5'-TGGTGGAGTGGGCTTCATG-3'
- RPL32-F: 5'-GACGCTTCAAGGGACAGTATCTG-3'
- RPL32-R: 5'-AAACGCGGTCTGCATGAG-3'

Ex vivo larval hemocyte phagocytosis assays

Ex vivo phagocytosis assays were performed using a mix of equal volumes of *E. coli* and *Staphylococcus aureus* AlexaFluorTM488 BioParticlesTM (Invitrogen), following manufacturer's instructions; and see (Melcarne et al., 2019 [↗](#)). Five L3 wandering larvae from our newly generated mutant lines, or carrying the Hml-Gal4, UAS-GFP hemocyte marker as a control, were bled into 150 μ L of Schneider's insect medium (Sigma-Aldrich) containing 1 μ M phenylthiourea (PTU; Sigma-Aldrich). The hemocyte suspension was then transferred to 1.5 mL low binding tubes

Drosophila lines	Reference	Name	Deficient pathway			
			IMD	Toll	Phagocytosis	Melanization
<i>iso ; iso ; Rel^{E20}</i>	Hedengren et al., 1999	Δ IMD	-	+	+	+
<i>iso ; iso ; spz^{mt7}</i>	Lemaitre et al., 1996	Δ TOLL	+	-	+	+
<i>iso ; NimC1¹ ; Eater¹</i>	Melcane et al., 2019	Δ Phag	+	+	-	+
<i>iso ; PPO1Δ, PPO2Δ; iso</i>	Bingelli et al., 2014	Δ MeI	+	+	+	-
<i>+ ; + ; Rel^{E20}, spz^{mt7}</i>	this paper	Δ IMD, Δ TOLL	-	-	+	+
<i>iso ; PPO1Δ, PPO2Δ ; NimC1¹, Eater¹</i>	this paper	Δ Phag, Δ MeI	+	+	-	-
<i>iso ; NimC1¹ ; Eater¹, Rel^{E20}</i>	this paper	Δ IMD, Δ Phag	-	+	-	+
<i>iso ; PPO1Δ PPO2Δ ; Rel^{E20}</i>	this paper	Δ IMD, Δ MeI	-	+	+	-
<i>iso ; PPO1Δ PPO2Δ ; spz^{mt7}</i>	this paper	Δ TOLL, Δ MeI ^{PPS}	+	-	+	-
<i>iso Hay-psh^{Def} ; iso ; iso</i>	Dudzic et al., 2019	Δ TOLL, Δ MeI ^{HP}	+	-	+	-
<i>iso Hay-psh^{Def} ; NimC1¹ ; Eater¹ Rel^{E20}</i>	this paper	Δ ITPM	-	-	-	-

Table 1

List of mutants used in this study.

"-" indicates the deleted immune module.

(Eppendorf, SigmaAldrich), and AlexaFluorTM488 bacteria BioParticlesTM were added. The samples were vortexed, incubated at room temperature for 2 hours to allow phagocytosis and then placed on ice to stop the reaction. The fluorescence of extracellular particles was quenched by adding 0.4% trypan blue (Sigma-Aldrich) diluted to 1/3 concentration. Phagocytosis was quantified using a flow cytometer (BD Accuri C6) to measure the fraction of cells phagocytosing, and their fluorescence intensity. Isogenic wild-type iso *w¹¹¹⁸* larvae and Hml-Gal4, UAS-GFP larvae with or without bacterial particles were used to define the gates for hemocyte counting and the thresholds for phagocytosed particle emission. The phagocytic index was calculated as follows:

$$\text{Fraction of haemocytes phagocytosing}(f) = \frac{[\text{number of haemocytes in fluorescence positive gate}]}{[\text{total number of haemocytes}]}$$

$$\text{Phagocytic index}(PI) = [\text{Mean fluorescence intensity of haemocytes in fluorescence positive gate}] \times f$$

Finally, due to experimenter differences in total hemocytes collected and knock-on effects to calculating phagocytic index, phagocytic index was normalized with the wild-type as a reference set to 100% within experiment batch.

Wounding experiment

Clean injury was performed with a needle sterilized with an EtOH and PBS wash. For imaging of the melanization reaction upon pricking, the thorax of 3-8 days old flies was pricked using a sterile needle (diameter ~0.1mm). Pictures were taken 24h post-pricking and categorized into normal, weak, or no melanization blackening seen at the injury site per Dudzic et al. (Dudzic et al., 2019 [DOI](#)).

Systemic infections and survival

Systemic infections were performed by pricking 3- to 5-day-old adult males in the thorax with a 0.1mm-thick needle dipped into a concentrated pellet of bacteria, yeast, or fungal spores. For natural infections with *A. fumigatus*, flies were anesthetized and then shaken on a sporulating plate of fungi for 30 seconds. For *B. bassiana*, flies were shaken for 30s in a tube containing an excess of commercial spores sufficient to coat all flies uniformly, of which the flies later remove the vast majority through grooming. At least two replicate survival experiments were performed for each infection, though in specific cases some genotypes may have been included in only one or no experiments. 20-35 flies were included per vial when possible, and maintained on cornmeal fly media containing (per 1L): 6.2g agar, 58.8g cornmeal, 58.8g yeast, 60mL grape fruit juice, 4.83mL propionic acid, and 26.5mL 100g/L moldex in pure ethanol. Survivals were scored daily, and flies were flipped to new food vials 3 times per week. Due to challenges working with genotypes of different viability, we could not include all genotypes in all experimental replicates. Moreover, comparisons across genotypes required multiple levels of consideration. Given the complexity of genotype-vs-reference-by-pathogen comparisons in our study, needing to compare infections both to clean injury within genotype, and to wild-type (WT, *DrosDel iso w¹¹¹⁸*) or *ΔITPM* (flies lacking all four immune modules) across treatments, we chose to highlight survival differences only when they were: **i**) consistent across experimental replicates; **ii**) of a consistent logic across comparable genotypes; for instance, compound mutants containing *ΔMel* (e.g. *ΔIMD*, *ΔMel*) should be as or more susceptible to infections as *ΔMel* alone if melanization is truly critical for defense; and **iii**) of a mean lifespan difference ≥1.0 days after accounting for comparisons with unchallenged or clean-injury data. Total experiments (N exp) and total flies per genotype are reported to provide summary statistics per pathogen-by-genotype interaction. The mean lifespan reported in survival summary data has a maximum of 7 or 10 days depending on experimental conditions. Kaplan-Meier survival curves are provided in the supplementary information.

Quantification of microbial load for growth kinetics

3-8 days old flies were infected with the indicated microbe and concentration per OD₆₀₀ as described in [Table S1](#), and allowed to recover. At the indicated time post-infection, flies were anesthetized using CO₂ and surface sterilized by washing them in 70% ethanol. Ethanol was removed, and then flies were homogenized using a Precellys bead beater at 6500 rpm for 30 s in LB broth (*Providencia rettgeri*), BHI (*S. aureus*), or YPG (*C. albicans*) with 500 µL as pools of 5 flies. These homogenates were serially diluted and 100 µL was plated on LB, BHI or YPG agar. Plates were incubated overnight, and colony-forming units (CFUs) were counted using a Interscience Scan 500 plate scanning colony counter, and validated independently by manual counts for a subset of plates to ensure accuracy. Statistical differences were tested using Brown-Forsythe and Welch ANOVA tests with Dunnett's multiple test correction, which consider unequal variance.

Results

A set of immune-deficient lines in the iso Drosdel background

Mutations affecting only one of the four main modules of the *Drosophila* systemic immune response have been described. These include *Relish*^{E20} (*Rel*^{E20}, deficient for the Imd pathway, referred to here as Δ IMD), *spatzle*^{rm7} (*spz*^{rm7}, deficient for the Toll pathway, referred to here as Δ TOLL), *PPO1*^Δ, *PPO2*^Δ (no melanization, referred to here as Δ Mel), and *NimC1*¹; *eater*¹ (reduced phagocytosis, referred to here as Δ Phag). In addition to blocking phagocytosis, Δ Phag flies have a defect in hemocyte sessility and a decreased number of hemocytes at the adult stage ([Bretscher et al., 2015](#); [Melcarne et al., 2019](#); [Melcarne, 2020](#)). We have preferred the use of *NimC1*¹; *eater*¹ to study the contribution of hemocytes to defense rather than other approaches with important indirect effects such as the pre-injection of latex beads to saturate phagocytes ([Elrod-Erickson et al., 2000](#)), or expression of pro-apoptotic genes in plasmatocytes ([Charroux and Royet, 2009](#); [Defaye et al., 2009](#)).

All the mutations in this study were isogenized for at least seven generations into the DrosDel iso w¹¹¹⁸ genetic background ([Ferreira et al., 2014](#); [Ryder et al., 2004](#)). We then generated by recombination six of the seven possible combinations of flies simultaneously lacking two immune modules, with the exception of Δ TOLL, Δ Phag as *NimC1*¹; *eater*¹, *spz*^{rm7} flies were not viable (see fly strains in [Table 1](#)). Of note, in our hands recombinant *Rel*^{E20}, *spz*^{rm7} flies were not viable in the DrosDel isogenic background, necessitating use of non-iso flies to assess host defense in Δ IMD, Δ TOLL flies. Over the course of our study, isogenic *NimC1*¹; *eater*¹ flies also displayed viability issues, and specific experiments used non-isogenic flies to improve confidence in phenotypes; these experiments are noted in figure captions. We used two approaches to simultaneously delete the Toll and melanization modules: **i)** *PPO1*^Δ, *PPO2*^Δ; *spz*^{rm7} flies (referred to as Δ TOLL, Δ Mel^{PPS}) and **ii)** flies carrying a small viable deletion *Hayan-psh*^{Def} (referred to as Δ TOLL, Δ Mel^{HP}) which removes two clustered serine protease genes, *Hayan*, and *persephone*, blocking the TOLL-PO SP cascade that regulates both humoral-Toll and melanization ([Dudzic et al., 2019](#); [Westlake et al., 2024](#)). This effectively deletes the Toll and melanization pathways using only one deletion event on the X chromosome. Excitingly, we succeeded in generating a fly line deficient for all four immune modules with the genotype: *Hayan-psh*^{Def}; *NimC1*¹; *eater*¹, *Rel*^{E20} hereafter named Δ ITPM (Δ IMD, Δ TOLL, Δ Phag, Δ Mel). These flies were homozygous viable (very poor viability), and fertile, despite major deficiencies in these four important systemic immune mechanisms. These flies allow us to monitor survival kinetics and pathogen growth in the near-total absence of an immune system. Finally, we added two iso fly lines in our analysis that delete key host defense peptides downstream of the Toll and Imd pathways: **i)** *Bom*^{Δ55C}, a short deletion removing ten Bomanins at position 55C, which causes a major susceptibility to infection by Gram-positive bacteria and fungi ([Clemmons et al., 2015](#)), and **ii)** Δ AMP14, a compound fly line of eight mutations removing 14 AMP genes (4 Cecropins, 4 Attacins, 2 Dipterocins, Drosocin, Defensin, Drosomycin, and

Metchnikowin), which causes a major susceptibility to infection by Gram-negative bacteria (Carboni et al., 2022 [↗](#); Hanson et al., 2019 [↗](#)). This set of immune-deficient fly lines in a defined genetic background provides a unique tool kit to systematically address the individual and collective contributions of immune modules to defense against a variety of pathogens.

Imd, Toll, Phagocytosis, and Melanization modules function largely independently

The Imd, Toll, Phagocytosis, and Melanization modules are interconnected (Westlake et al., 2024 [↗](#)). For instance, the Toll and Imd pathways both regulate a subset of genes involved in melanization (De Gregorio et al., 2002 [↗](#); Ligoxygakis et al., 2002 [↗](#)). Moreover, one module can indirectly affect another. For instance, a lack of melanization could increase the bacterial load after infection, resulting in higher activation of the Toll and Imd pathways (Binggeli et al., 2014 [↗](#)). In the first step, we analyzed the activation of each of the four modules in single- and multiple-module deficient flies using classical readouts of these modules. This also allowed monitoring of the possible effects of deleting one module on the activity of another. We analyzed the immune response in single module deficient flies by monitoring **i)** the expression of the antibacterial peptide gene *Diptericin* (*DptA* - a target of the Imd pathway), and **ii)** the antifungal peptide gene *Drosomycin* (*Drs* - a target of the Toll pathway) upon systemic infection with a mixture of heat-killed *Escherichia coli* and *Micrococcus luteus*, **iii)** ex vivo phagocytosis of *E. coli* and *Staphylococcus aureus* coated bioparticles incubated with larval hemocytes, and **iv)** melanization at the injury site of adult flies. This analysis first confirmed that Δ IMD, Δ TOLL, Δ Phag, and Δ Mel flies are deficient for Imd, Toll, phagocytosis, and melanization respectively (Figure 1A-D [↗](#)). We note that the expression level of *DptA* was roughly similar to the wild-type in Δ TOLL and Δ Mel flies, however, Δ Phag flies show reduced *DptA* induction at 6h compared to WT (~50%) (Figure 1A [↗](#)). This represents an immense induction of *DptA* after infection, but may suggest either a lag or lesser overall Imd response in Δ Phag flies. *DptA* expression in Δ Phag was, however, comparable to WT at 12h and 24h. On the other hand, *Drs* induction at 6h in Δ Mel flies was significantly higher than WT (~1.6x), as found previously using live *M. luteus* infection (Binggeli et al., 2014 [↗](#)) (Figure 1B [↗](#)). The phagocytic assay confirmed that the Δ Phag mutant was significantly impaired in its ability to phagocytose bacterial bioparticles. Furthermore, Δ IMD, and Δ Mel mutant larvae had a phagocytic index similar to WT (Figure 1C [↗](#)), while Δ TOLL had a higher phagocytic index, but with high variance. As expected, there was no apparent melanization at the site of the injury in Δ Mel flies while blackening was readily observed in the wild-type and in other single-module mutants (Figure 1D [↗](#)). Collectively, this analysis indicates that each of the four immune modules can be selectively blocked in the isogenic DrosDel background without interfering with the competence of the others. We did note in some cases a modest change in module activity in the absence of another (higher Toll activity in Δ Mel, lower IMD activity in Δ Phag, and higher phagocytic activity in Δ TOLL), findings that require further exploration to establish consistency.

We then repeated our analysis using our set of module compound mutants. Unsurprisingly, all compound flies displayed the expected immune deficiencies, validating the lines we constructed. This analysis confirms that Δ ITPM flies, although viable, are indeed fully immune deficient for the four host defense modules. We did not detect any striking synergistic or antagonistic effects where deleting two modules had a vastly different impact compared to the additive effect of deleting the modules individually. Notably, a small level of *Drs* expression is retained in Δ TOLL mutant flies, but was fully abolished in Δ IMD, Δ TOLL flies (Figure 1A [↗](#) vs 1E [↗](#)), confirming observations that the *Drs* gene receives a small input from the Imd pathway (Leulier et al., 2000 [↗](#)). The Δ TOLL, Δ Mel^{PPS} and, Δ TOLL, Δ Mel^{HP} flies display *Drs* induction at 6h that is resolved to unchallenged WT levels by 12h, an expression pattern consistent with mediation of this induction by early input from the IMD pathway in these backgrounds. Taken together, all the compound mutant flies behave largely as expected given the sum effects of their individual mutations, confirming that these modules largely function independently.

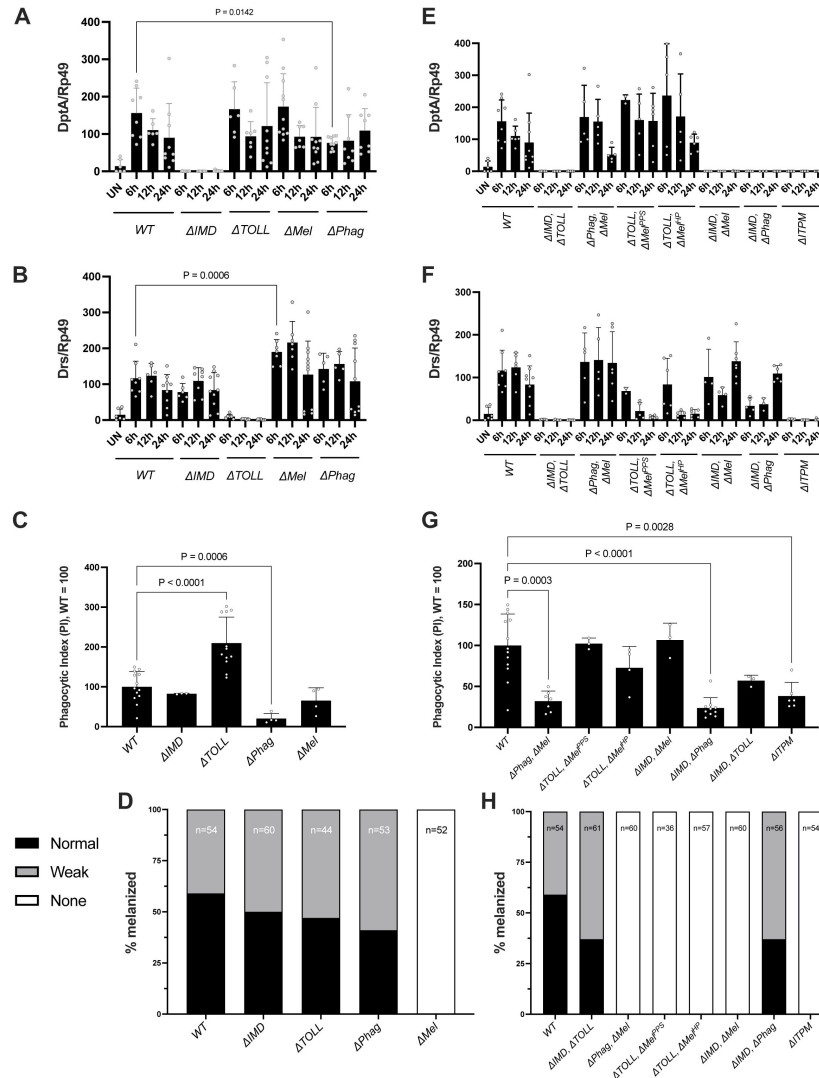


Figure 1

Each immune module functions largely independently from other immune modules.

A) Activation of the Imd pathway, as revealed by *DptA* expression following infection with a mixture of heat-killed *E. coli* and *M. luteus*, is broadly wild-type (WT) in single module mutants other than Imd. In the case of Δ Phag flies, we observed a lesser induction at 6h compared to wild-type. **B)** Activation of the Toll pathway as revealed by the expression of *Drs* is broadly wild-type in single module mutants other than Δ TOLL flies. Note that *Drs* receives a minor input from Imd signaling (Leulier et al., 2000), explaining minor induction of *Drs* in Δ TOLL flies at early time points. In the case of Δ MeI, we observed a slightly greater Toll pathway activity. **C)** The ability of plasmatocytes to phagocytose bacterial particles is not negatively affected in single module mutants except Δ Phag (also see Figure 1supp1), **D)** Cuticle blackening after clean injury is not impaired except in Δ MeI flies. **E)** Activation of the Imd pathway remains strongly inducible in compound mutants except when deficient for the Imd pathway. **F)** Activation of the Toll pathway remains strongly inducible in compound mutants except when flies were deficient of the Toll pathway. **G)** The ability of plasmatocytes to phagocytose bacterial particles is not significantly affected in compound mutants except those including Δ Phag. **H)** Cuticle blackening after clean injury is not impaired in compound mutants except in those including Δ MeI.

Both the Melanization and TOLL modules contribute to lifespan and survival to wounding

Before assessing resistance to infection, we monitored fly survival at 25°C and 29°C to determine if module mutations impact viability in the absence of challenge. At 25°C, no single module mutants died significantly more than wild-type in the absence of challenge within the ten-day window (**Figure 2A**). However, even in the absence of challenge all compound mutants other than ΔIMD , $\Delta TOLL$ suffered at least 25% cumulative mortality by 10 days post-eclosion, while $\Delta TOLL$, ΔMel^{PPS} and $\Delta ITPM$ flies suffered even greater cumulative mortality. Similar trends were observed at 29°C with slightly elevated mortality (**Figure 2supp1**). The observation that $\Delta TOLL$, ΔMel^{PPS} ($PPO1^1, PPO2^1, spz^{rm7}$) flies have shorter lifespan compared to $\Delta TOLL$, ΔMel^{HP} (*Hayan-psh^{Def}*) could be explained by residual Toll and/or PO activity in *Hayan-psh^{Def}* or additional indirect effects in $\Delta TOLL$, ΔMel^{PPS} flies. As such, the combined effect of Toll and Melanization deficiencies in unchallenged and clean injury survival is best assessed by considering signals from both genotypes. $\Delta ITPM$ flies (*Hayan-psh^{Def}*; *NimC1¹*; *Eater¹*, *Rel^{E20}*) have the lowest unchallenged survival rate of genotypes included in this study. Taken together, the Imd, Toll, Melanization, and Phagocytosis modules contribute little to lifespan maintenance individually, but are crucial to maintain lifespan collectively (**Figure 2B**).

Systemic infections were performed by septic injury (pricking with a contaminated needle). We therefore monitored the survival of single and double mutant flies after a clean injury to disentangle the effects of injury from those of infection. We found a cumulative mortality of ~50% for ΔMel flies by day 10 (**Figure 2C**), while other single module mutants displayed relatively little mortality over the 10-day window; ΔIMD had notable late-onset mortality in a minority of experimental replicates, which may be due to stochastic dysbiosis effects during aging (Hanson and Lemaître, 2023; Marra et al., 2021). Double mutant ΔIMD , $\Delta TOLL$ and ΔIMD , $\Delta Phag$ flies retained a high survival rate. Consistent with an importance of melanization in the injury response (Binggeli et al., 2014), all the compound mutants affected in melanization were susceptible to clean injury, with similar mortality seen in $\Delta Phag$, ΔMel , ΔIMD , ΔMel , $\Delta TOLL$, ΔMel^{PPS} , $\Delta TOLL$, ΔMel^{HP} , and $\Delta ITPM$ flies. Thus, while these genotypes differ in their base lifespan, their viability upon injury is remarkably similar. These results emphasize the interconnectedness of the melanization response with every other immune module for maintaining health after injury.

Similar results were obtained for both unchallenged and clean injury lifespans at 29°C. Of note, $\Delta TOLL$, ΔMel^{PPS} suffered significantly greater mortality in unchallenged conditions than other genotypes, even including $\Delta ITPM$ flies (**Figure 2supp1**). Other compound mutant genotypes displayed a continuum of mortality in unchallenged conditions between $\Delta ITPM$ (~50% cumulative mortality) and $\Delta TOLL$, ΔMel^{HP} (~15% cumulative mortality). Upon clean injury, results were broadly consistent with 25°C trends, with a more uniform mortality across all genotypes.

Collectively, our study highlights that all these immune modules contribute somewhat to the survival of flies upon clean injury, with a major role of the melanization response in maintaining lifespan and survival to injury. In the next steps of our study, we compared survival upon infection to both unchallenged and clean injured flies to distinguish the impact of microbial infection from the injury itself.

Multiple mechanisms contribute to resistance to microbial infection

To compare the contribution of each module to host defense, we performed infection experiments of wild-type and single- and multiple-module flies following septic infection with five viruses, eight Gram-positive bacteria, eight Gram-negative bacteria, septic injury for two fungal species,

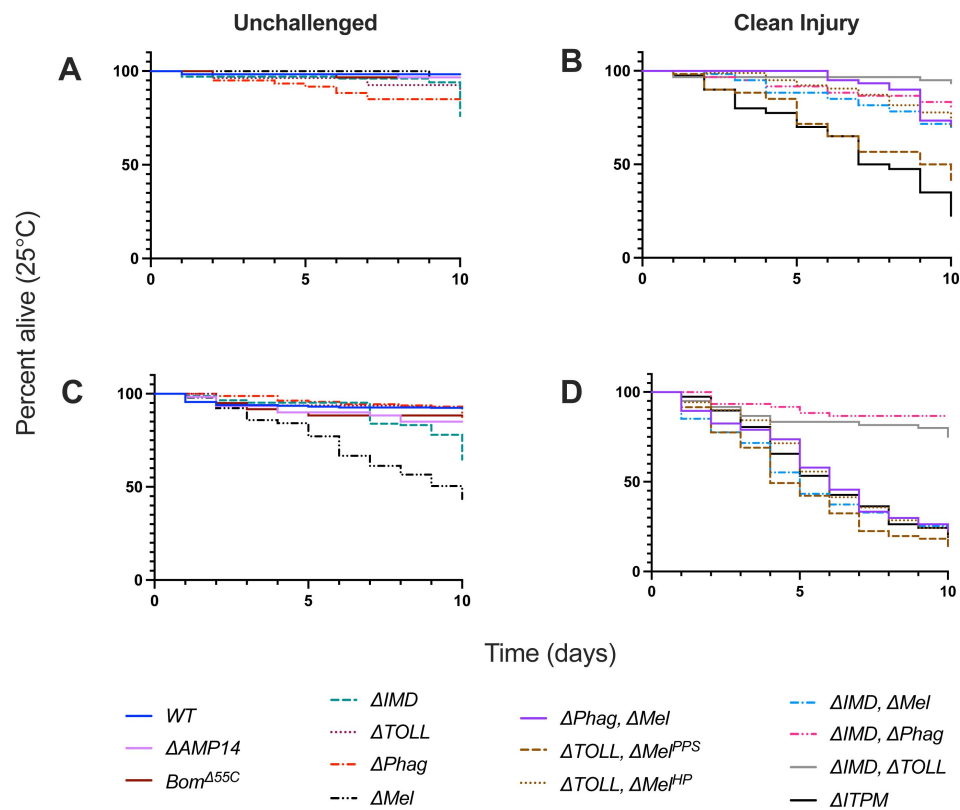


Figure 2

Lifespans of mutants used in this study in unchallenged flies and upon clean injury conditions at 25°C.

A-B) Single module mutants; **C-D)** compound module mutants. See **Figure 2supp1** [for 29°C comparisons](#).

and natural infection (*i.e.* spore deposition on the cuticle) for two fungal species. Temperature, dose, and days monitored are shown in [Figure 3](#), and see [Table S1](#) for other key parameters. Survival curves are shown in [Supplementary file 1](#) for each pathogen and results of these survivals are summarized in [Figure 3](#).

We performed infection experiments with 24 viruses and microbial species: DCV, FHV, DXV, IIV-6, SINV, *Aspergillus fumigatus*, *Beauveria bassiana*, *Candida albicans*, *Mycobacterium marinum*, *Corynebacterium diphtheriae*, *Micrococcus luteus*, *Streptococcus pneumoniae*, *Enterococcus faecalis*, *Staphylococcus aureus*, *Listeria monocytogenes*, *Bacillus subtilis*, *Vibrio parahaemolyticus*, *Providencia rettgeri*, *Providencia burhododranaria*, *Pectobacterium carotovorum* Ecc15, *Klebsiella pneumoniae*, *Enterobacter cloacae*, *Escherichia coli*, and *Salmonella enterica ser. typhimurium*. Collectively, this represents infections with 24 pathogens across 14 genotypes and two infection routes (~350 genotype-by-pathogen interactions). When combined with comparisons to clean injury and unchallenged controls to validate if a lifespan difference is meaningful, and to wild-type or $\Delta ITPM$ treatments, this inflates to 1000+ interactions. We therefore focused our survival analysis on major effects. We report the core summary statistics and overt trends, and comment only where difference in mean lifespan between treatments and reference genotypes or controls was a minimum of 1 day.

Our study confirms that the Imd pathway plays a major role against all tested Gram-negative bacteria, as ΔIMD flies succumb more rapidly than wild-type flies to all of them. This pathway also contributes, albeit to a lesser extent, to survival against the Gram-positive bacteria *B. subtilis* and *L. monocytogenes*, which bear DAP-type peptidoglycan at their membrane that triggers Imd activation ([Kaneko et al., 2004](#); [Leulier et al., 2003](#)). We were interested to see the impact of our ΔIMD flies in host defense to viruses as the specific mutation we used (*Rel^{E20}*) deletes a transcription factor common to both the canonical Imd pathway and the recently-appreciated cGLR-Sting-Relish pathway with roles in viral defense ([Goto et al., 2018](#)). We found that ΔIMD mutant flies were somewhat susceptible to the four viruses DCV, FHV, DXV, and IIV6, but not SINV, agreeing with previous studies showing a reduction in lifespan of 1-2 days for *Rel* mutant flies after viral infection ([Costa et al., 2009](#); [Goto et al., 2018](#); [Sansone et al., 2015](#)).

This study also confirms a prominent role of the Toll pathway in survival upon infection by all virulent Gram-positive bacteria and fungi upon septic injury. Less virulent microbes and infectious routes (such as *A. fumigatus* by natural infection) did not result in increased mortality in individual $\Delta TOLL$ mutant flies. A contribution of the Toll pathway to survival was also observed for some Gram-negative bacteria, including a minor contribution to survival upon *P. rettgeri* infection, and a major contribution after *V. parahaemolyticus* infection. In addition, we found a susceptibility of $\Delta TOLL$ flies to FHV, DXV, and IIV6, suggesting Toll-regulated effectors mediate defense against some viral infections.

Use of $\Delta Phag$ mutants reveals a role of the cellular response in survival against the fungus *B. bassiana*. $\Delta Phag$ flies further show a minor susceptibility to *S. aureus*, and a subset of Gram-negative bacteria (*P. rettgeri*, *Ecc15*, and *E. coli*). Susceptibilities to Gram-negative bacteria could result from the observed lag in Imd pathway activation ([Figure 1A](#)); indeed, previous studies have shown a full *DptA* response is pivotal for defense against *P. rettgeri* ([Hanson et al., 2023](#), [2019](#); [Unckless et al., 2016](#)), and lifespans are greatly impacted by subtle difference for pathogens with intermediate levels of virulence ([Duneau et al., 2017](#)). A role for Imd in explaining $\Delta Phag$ susceptibilities does not detract from the importance of the cellular response, but instead offers a route to explore mechanism for these phenotypes.

Surprisingly, our study reveals a relatively consistent role of melanization against all the tested viruses, notably so for DXV, IIV-6, and SINV. ΔMel flies also display a minor increased susceptibility to *B. bassiana* natural infection and septic injury. As previously described ([Dudzic et al., 2019](#)), ΔMel flies are highly susceptible to the Gram-positive bacterium *S. aureus*, rivaling the

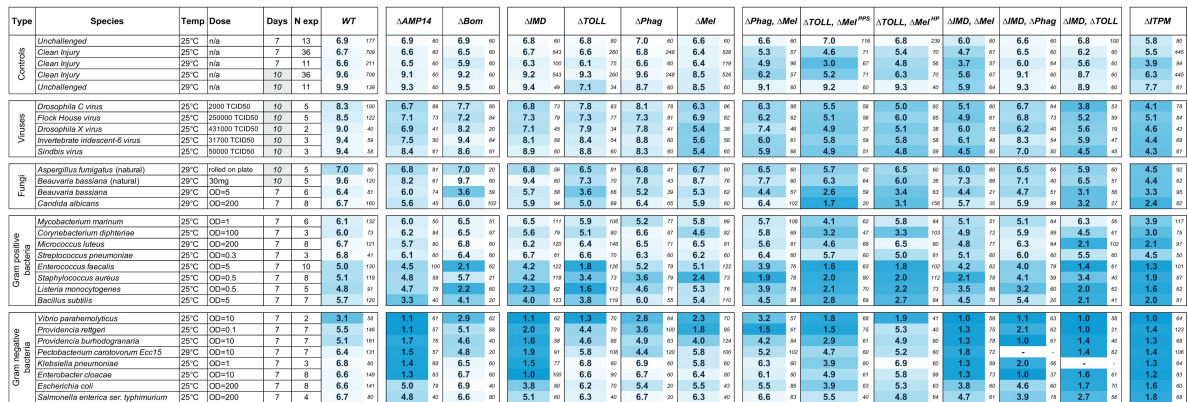


Figure 3

Heatmap of lifespans of immune module-deficient flies upon infection by various pathogens..

Darker blue indicates lower survival, while white indicates maximum survival. Experiments used either 7 or 10 days as a maximum lifespan/time course, and the heatmap is adjusted accordingly per row to have white fill for the maximum possible lifespan of that row. Small text to right of mean lifespan indicates total flies. Survival curves are presented in **Supplementary file 1**, and a summary of susceptibilities is provided in **Figure 3supp1**.

susceptibility of $\Delta ITPM$ flies. ΔMel flies further show a strong susceptibility to *P. rettgeri* rivalling ΔIMD flies, and also a minor susceptibility to infection by other Gram-negative bacteria (*P. burhodogranariae*, *E. coli*).

Collectively, the use of single module mutants confirms the major role of the Imd pathway against Gram-negative bacteria and Toll against virulent Gram-positive bacteria and fungi. Interestingly, melanization was consistently important to survive viral infection, and plays a prominent and important role in defense against specific bacterial species. While it was almost never the most critical module, phagocytosis and the cellular response contributes to survival against various germs. Use of single module deficient flies alongside $\Delta ITPM$ flies reveals that survival to some of the microbes tested relies almost entirely on one pathway (Imd – *Ecc15*, *K. pneumoniae*, *E. cloacae*, *P. burhodogranariae*; Toll – *E. faecalis*), or can receive notable contributions from two pathways (Toll and IMD – *L. monocytogenes*, *B. subtilis*, *V. parahaemolyticus*; Toll and Phagocytosis – *A. fumigatus* and *B. bassiana* natural infection; Imd and Melanization – DCV). Strikingly, many pathways contribute to host defense to *S. aureus* and *P. rettgeri*.

Contribution of AMP and Bomanins to Toll and Imd activities

Recent studies have described a prime importance of antimicrobial peptides (AMPs) and host defense peptides (HDPs) in responding to infection (Clemmons et al., 2015; Hanson, 2024; Hanson et al., 2019). These peptides are the principal effectors of the Toll and Imd pathways, yet mutations for AMPs and HDPs typically have not been tested alongside mutations for both the Imd and the Toll pathways across pathogens, as studies have typically included just the single relevant pathway mutation as a positive control. Thus, the importance of AMPs and HDPs relative to the pathways that regulate them has been only partially investigated. We used $\Delta AMP14$ and ΔBom flies to assess the extent to which AMPs and Bomanins mediate Toll and Imd contributions to defense (Figure 3). Here we found that AMPs are critical for survival to infection against all Gram-negative bacteria tested, often rivaling the susceptibility of ΔIMD flies. Notably, $\Delta AMP14$ flies were also somewhat susceptible to the yeast *C. albicans*, and the Gram-positive bacteria *B. subtilis* and *M. luteus*, reinforcing minor susceptibilities seen previously (Carboni et al., 2022; Hanson et al., 2019). Interestingly, we found that $\Delta AMP14$ survivals to DCV, FHV, DXV, and IIV6, largely paralleled those of ΔIMD flies (Figure 3, Supplemental File 1), suggesting Imd-mediated survival phenotypes rely greatly on the presence of the seven classical AMP families.

On the other hand, ΔBom flies were susceptible to septic injury by *B. bassiana* and also somewhat to *C. albicans*, consistent with previous studies (Supplementary file 1), as well as all virulent Gram-positive bacteria, consistent with their important role downstream of the Toll pathway (Clemmons et al., 2015; Lindsay et al., 2018; Xu et al., 2023). In addition, ΔBom flies paralleled the $\Delta TOLL$ susceptibility to DCV, FHV, and DXV, but were not susceptible to IIV6 like $\Delta TOLL$.

Collectively, our study confirms the key role of antimicrobial peptides downstream of the Imd pathway in the defense against Gram-negative bacteria and a primary role of Bomanins downstream of Toll in defense against fungi and Gram-positive bacteria, with little input to survival against Gram-negative bacteria. We additionally find these secreted peptides can contribute to survival after certain viral infections.

Immune modules mostly contribute additively to host defense

The results described above reveal the key contribution of single modules and immune effectors to host defense. However, they did not assess possible additive, synergistic, or antagonistic contribution of these four modules. Moreover, they did not monitor to what extent these modules are important in the absence of other modules. To address this, we compared the survival of single-module deficient flies to flies lacking two modules, and $\Delta ITPM$ flies lacking all four modules (Figure 3).

We first observed that $\Delta ITPM$ flies, which lack the four modules, were always as or more susceptible than single- and double-module deficient flies. In multiple microbe-specific cases, single or double mutants rivalled the susceptibility of $\Delta ITPM$ flies. In particular, the susceptibility of ΔIMD -flies to most Gram-negative bacteria rivalled $\Delta ITPM$, and for some germs $\Delta TOLL$ (*L. monocytogenes*, *E. faecalis*, *B. bassiana* injury) or ΔMel (*S. aureus*) alone was comparable to $\Delta ITPM$. In other cases, however, combined loss of two pathways was necessary to rival $\Delta ITPM$ susceptibilities. For instance, ΔIMD , $\Delta TOLL$ deletion causes complete $\Delta ITPM$ -like susceptibility to *B. subtilis*, *M. luteus*, and *E. coli*, and near-complete susceptibility to *S. typhimurium*. ΔIMD , $\Delta TOLL$ further causes increased susceptibility to most viruses compared to its single module counterparts. In these cases, there is clear synergy or additivity of the contributions of the two modules, as either pathway alone results in only minor or no susceptibility. We also observed intriguing differences between our two versions of $\Delta TOLL$, ΔMel flies, as $\Delta TOLL$, ΔMel^{PPS} tended to succumb to infection to a greater extent than $\Delta TOLL$, ΔMel^{HP} , consistent with the differences observed upon clean injury (**Figure 2**). The $\Delta TOLL$, ΔMel^{PPS} displayed a minor increase in susceptibility compared to $\Delta TOLL$, ΔMel^{HP} to *A. fumigatus* natural infection, and more prominent susceptibilities to the yeast *C. albicans*, the Gram-positive bacteria *M. marinum*, *M. luteus* and also to Gram-negative bacteria *K. pneumoniae*, *E. coli*, *P. rettgeri* and *P. burhododranariae*.

Collectively, use of double-module deficient flies uncovers contributions of immune modules to host defense that were masked by other modules. While ΔIMD and $\Delta TOLL$ deficiencies sometimes displayed a synergistic effect (*i.e.* little mortality of single mutants, but $\Delta ITPM$ -like mortality of double mutants), for combinations of other pathways the results were more often minor with a cumulative effect on susceptibility, suggesting most modules contribute to defense independently. The use of two variations of $\Delta TOLL$, ΔMel , with $\Delta TOLL$, ΔMel^{PPS} having a stronger impact than $\Delta TOLL$, ΔMel^{HP} , also highlights how the level of pathway disruption can affect infection outcomes, particularly for certain fungal or bacterial pathogens like *Providencia* species that may cleave host immune proteases (suggested by [Duneau et al., 2017](#); [Issa et al., 2018](#)).

Timing of Toll, Imd, Phagocytosis, and Melanization contribution to host defense

Host defense programs can rely on resistance mechanisms that directly target or limit growth of pathogens ([Howick and Lazzaro, 2017](#); [Lafont et al., 2021](#); [Medzhitov et al., 2012](#)). When effectors reach a critical concentration threshold that promotes resistance, pathogen growth is inhibited; for instance, the time to pathogen control of *P. rettgeri* is ~7 hours in *Drosophila*, relying heavily on the expression and production of DptA ([Duneau et al., 2017](#); [Hanson et al., 2023](#)). Phagocytosis and the melanization response are often described as providing near-immediate immune protection ([Haine et al., 2008](#)), with reactions *ex vivo* progressing within minutes. Meanwhile activation of the Imd and Toll pathways takes hours to reach peak concentrations of host defense peptides ([Uttenweiler-Joseph et al., 1998](#)). Thus, we were curious if different mutations would alter time of action to control microbial growth. We therefore measured the microbial growth rate for *P. rettgeri*, *S. aureus*, and *C. albicans* at different time points post-infection in wild-type and single module mutant flies as these pathogens were combatted by several immune modules in our survival data.

Agreeing with contributions to survival, flies singly deficient for IMD, Phagocytosis, or Melanization modules display a higher bacterial load than wild-type when infected with *P. rettgeri* (**Figure 4A**), which was apparent at 12 hours post-infection. Our observation that $\Delta Phag$ showed impaired IMD signaling at 6h (**Figure 1A**) may explain the contribution of phagocytosis to *P. rettgeri* microbial growth control with the same timing as ΔIMD flies. We observed that ΔIMD flies, ΔMel flies, and $\Delta ITPM$ flies, each of which succumb completely to *P. rettgeri* infection, also show high and consistent microbe loads at 24h that are indicative of sepsis-induced death. On the other hand, wild-type, $\Delta TOLL$, and $\Delta Phag$ flies show a greater stochasticity in bacterial load at 24h that is

consistent with a fraction of individuals surviving the infection by controlling the pathogen. This indicates deletion of the Toll pathway and phagocytosis does not fully ablate resistance mechanisms essential for combatting *P. rettgeri*.

In the case of *S. aureus* (**Figure 4B** [↗](#)), we observed some individuals with higher bacterial loads in Δ Phag, and Δ IMD flies at 2h and 6h, which was not seen in wild-type, Δ TOLL, or Δ Mel flies. This suggests that the Imd pathway can suppress early *S. aureus* growth, and possibly also phagocytosis, although this difference in Δ Phag could ultimately stem from reduced IMD induction. Interestingly, flies lacking melanization displayed the highest susceptibility, but did not depart from wild-type or other modules in pathogen load until 12h post-infection. While the melanization reaction is a rapid response, these results suggest that the killing activity of the melanization response acts with slower kinetics. In Δ ITPM flies, we observed earlier *S. aureus* growth at 2h and 6h as in Δ IMD and Δ Phag flies. However, *S. aureus* loads at 12-24h were more comparable in Δ ITPM flies to Δ Mel flies. These trends agree both with survival data kinetics, and an independent action of each of these pathways in their resistance effects against *S. aureus*. This suggests that Δ ITPM flies succumb even more quickly than individual module mutants due to loss of multiple resistance mechanisms with different kinetics of activation that independently contribute to defense.

For *C. albicans* (**Figure 4C** [↗](#)), mortality begins 2-3 days post-infection in Δ TOLL mutants, but takes place at later time points for Δ IMD, Δ Phag, and Δ Mel flies. Microbial growth kinetics show high *C. albicans* loads at 24h in Δ TOLL, consistent with a previous study (Hanson et al., 2019 [↗](#)) and observations done with *Candida glabrata* (Quintin et al., 2013 [↗](#)). Interestingly, a few Δ IMD flies also showed elevated growth of *C. albicans* at early time points. This suggests Imd-regulated effectors may help to suppress initial *C. albicans* growth, while Toll-regulated genes (e.g. Bomanins, Drs) contribute to this pathogen suppression to a greater extent at later time points. On the other hand, Δ Phag and Δ Mel flies did not show any increased *C. albicans* load within the 24h time window, despite onset of mortality at later time points. Finally, Δ ITPM flies displayed more consistent and rapid growth of *C. albicans* than Δ TOLL or Δ IMD alone, particularly visible at 12-24h post-infection, consistent with independent contributions of both Toll and Imd in resistance to this yeast.

Collectively, we observed that microbial growth in each module mutant parallels susceptibility, indicating that they contribute to resistance. Surprisingly we did not observe an early role in pathogen killing for melanization despite the rapid blackening response mediated by melanization in ex vivo assays (Nakhleh et al., 2017 [↗](#)). We further recover stochasticity in microbe loads across genotypes, as expected when only a subset of individuals suppress the pathogen and survive the infection, resulting in bimodal outcomes (Duneau et al., 2017 [↗](#)). The more rapid and consistent microbial growth kinetics of Δ ITPM flies, particularly visible at 18h, further demonstrate that the modules contribute to resistance collectively.

Contributions of individual modules to disease tolerance

A metric increasingly used to delineate roles of resistance and tolerance is the Pathogen Load Upon Death (PLUD) (Duneau et al., 2017 [↗](#); Duneau et al., 2025 [↗](#)). PLUD is affected by both the virulence of the pathogen and tolerance of the host to survive to a given pathogen burden before succumbing to infection. We monitored PLUD for the same pathogens used to determine the timing of defense modules: *P. rettgeri*, *S. aureus*, and *C. albicans*. The PLUD of both *P. rettgeri* and *S. aureus* was overall similar in wild-type and single-module mutant flies (**Figure 5A-B** [↗](#)). Notably, there was a greater stochasticity in *P. rettgeri* infections trending towards lower PLUD in Δ ITPM flies, suggesting a stochastic reduction in disease tolerance of Δ ITPM flies to this bacterium, although this was not significant compared to wild-type ($P > .05$).

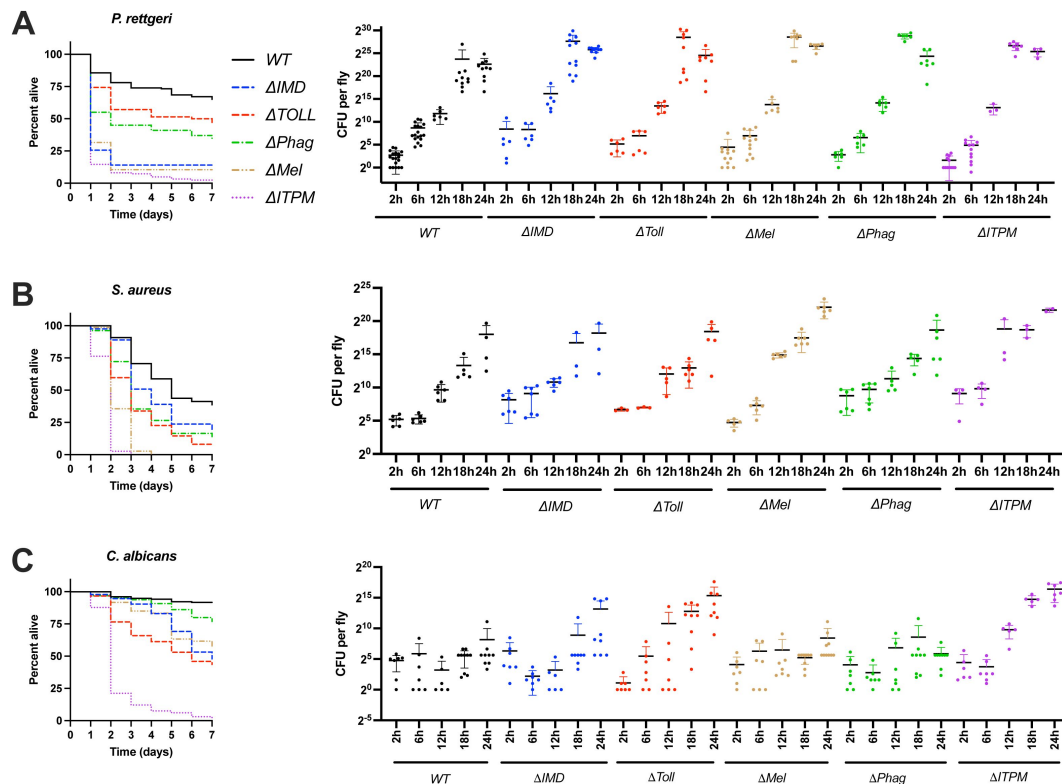


Figure 4

Growth kinetics of *P. rettgeri* (A), *S. aureus* (B), and *C. albicans* (C) in wild-type, single module mutant and $\Delta ITPM$ flies..

Survival curves from [Supplementary file 1](#) underlying data in [Figure 3](#) are shown for context. Each data point reflects a pooled sample of 5 flies.

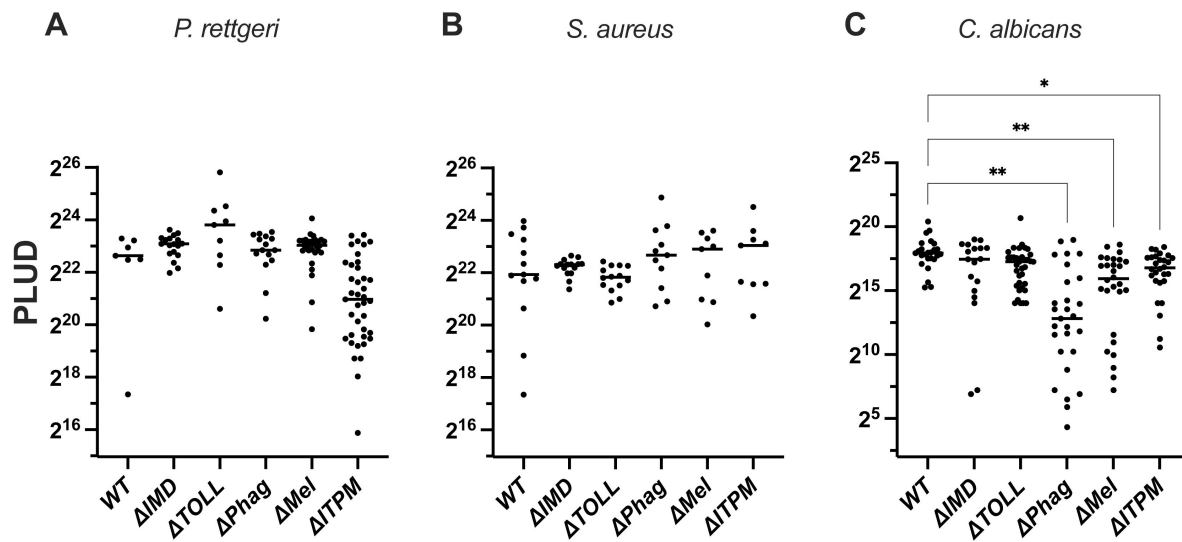


Figure 5

Measurement of PLUD upon infection with *P. rettgeri*, *S. aureus*, and *C. albicans*.

A) The PLUD of *P. rettgeri* in individual module mutant flies is not significantly different from wild-type. The PLUD of *ΔITPM* flies was not significantly different, although censoring of a single low-PLUD outlier in the wild-type would result in a significant difference between wild-type and *ΔITPM* flies ($p < .01$). **B)** The PLUD of *S. aureus*-infected flies is not different across any genotype. **C)** The PLUD of *C. albicans*-infected flies was significantly lower in *ΔPhag* flies with a notably lower mean PLUD. This difference was robust to use of different *ΔPhag* genetic backgrounds (merged data shown here). *ΔMel* flies and *ΔITPM* flies also had significantly lower PLUD. *, $P < 0.05$; **, $P < 0.01$

For *C. albicans*, the distribution of PLUD values for $\Delta Phag$ flies was markedly different from wild-type, also seen to some extent for ΔMel . Interestingly, flies with these genotypes suppress *C. albicans* growth well (**Figure 4C**). Yet here we recovered many individuals from both of these module-deficient lines that died with a far lower PLUD (**Figure 5C**). Taken together, this suggests that the humoral immune pathways, particularly Toll, contribute to resistance against *C. albicans*. However, the cellular response and melanization reaction instead regulate tolerance to *C. albicans* infection. The use of $\Delta ITPM$ flies further emphasizes that when both resistance (Toll, Imd) and tolerance (Phagocytosis, Melanization) mechanisms are deficient, the loss of resistance plays a primary importance, and can mask observations of reduced tolerance.

Collectively, monitoring of PLUD suggests that some modules, specifically phagocytosis and melanization, contribute to tolerance of certain infections. Importantly, this tolerance effect was only seen in genotypes that are able to resist infection. These findings demonstrate the collective contribution to resistance and/or tolerance of the four immune modules studied here.

Discussion

In this article, we have generated a set of single and compound immune module deficient fly lines in a controlled genetic background. Using various assays, we confirmed the validity of our lines, revealing that each module can be activated independently. We did, however, observe higher Toll activation in ΔMel flies upon systemic infection with dead bacteria. Future studies may reveal that this higher Toll activity in ΔMel flies is due to interaction with a regulatory pathway, or reflects higher persistence of microbial elicitors (e.g. peptidoglycan) that activate the Toll pathway in melanization-deficient flies. We also observed reduced Imd activation in $\Delta Phag$ flies at 6 hours post-infection, which could reflect a role for hemocytes in stimulation of the Imd systemic response. In this study, we used *NimC1¹; eater¹* double mutants to assess the cellular response. These flies have defects in phagocytosis, but also hemocyte adhesion and sessility (Melcarne et al., 2019). *NimC1¹; eater¹* larvae also have increased hemocyte number at the larval stage, but Hml positive cells are rapidly lost in adults (Melcarne, 2020). Despite these limitations, we believe that the mutations we have used here are among the best available to assess the roles of these four modules to host defense. Surprisingly, we were able to produce a fly line lacking all four main defense mechanisms of the systemic immune response, indicating that none of these modules is essential for survival. Theoretically, $\Delta ITPM$ flies are almost completely immune deficient, but they are still able to clot wounds, activate the JNK and JAK-STAT pathways that mediate the wound healing response, and retain constitutive immune defense molecules that could provide a certain degree of protection. Consistent with several studies (Capilla et al., 2017; Carvalho et al., 2014; R  met et al., 2002), we show that Toll and melanization contribute synergistically to wound healing in adults. However, the observation that $\Delta Phag$, ΔMel or ΔIMD , ΔMel double mutant flies are also more susceptible to clean injury than ΔMel flies indicates that the Imd pathway and phagocytosis also contribute to wound healing. Thus, although immune deficient fly lines for the four modules are viable, the use of compound module mutants and $\Delta ITPM$ flies reveals a clear role of the immune system in response to wounding and lifespan maintenance.

Use of single- and double module-deficient flies provides key insights on the mechanisms used by *Drosophila* to combat infection. We confirm previous studies revealing the role of Imd against Gram-negative bacteria, Toll against Gram positive bacteria and virulent fungi, and an importance of Melanization and Phagocytosis against specific pathogens (Charroux and Royet, 2009; Defaye et al., 2009; Garg and Wu, 2014; Nehme et al., 2011). Surprisingly, the Melanization module was consistently important in survival to viral infection. It seems unlikely that prophenoloxidase activity in the hemolymph can combat intracellular viral agents. We instead speculate this contribution of melanization to viral defense could be due to its role in wound healing, or perhaps autotoxic contributions of melanization reaction intermediates that fail to convert in phenoloxidase-deficient flies. We additionally recovered a role of Imd signaling (i.e. *Relish*) in

antiviral defense, which was expected given recent characterizations of cGIR-Sting-Relish antiviral immunity (Ai et al., 2024; Cai et al., 2022; Goto et al., 2018). However, susceptibilities of *Rel^{E20}* flies were paralleled by Δ AMP14 in all cases, suggesting AMPs largely mediate this susceptibility. While Sting regulates a number of genes likely important for antiviral defense (Goto et al., 2018), the susceptibility we observe here could be a direct action of AMPs on enveloped viruses as described in some studies (Feng et al., 2020; Huang et al., 2013; Yasin et al., 2004), or an indirect effect, such as a need to regulate gut microbes after the damage induced by viral replication (Marra et al., 2021). We additionally used dual modes of infection for the fungus *B. bassiana*, finding that both Melanization and Toll modules have a role in both infection modes. However, our study reveals that Bomanin effectors explain most of the Toll contribution upon septic injury but not natural infection by *B. bassiana*. This observation is in line with previous observations showing differential importance of effectors or modules according to infection route (Martins et al., 2013).

Our double module mutant analysis reveals that most modules contribute additively to host defense. This is consistent with largely independent function of these modules. However, we noted some instances of synergy between two pathways, notably Toll and Imd. Synergy between Toll and Imd can be explained by the fact that many immune-inducible regulated genes receive input of both the Imd or Toll pathways; some like *Metchnikowin*, *Drosomycin* and *Transferrin1* that can be induced through either Toll or Imd (De Gregorio et al., 2002). Finally, we observed rarer cases of synergy between Toll and Melanization, or Imd and Melanization.

Our study confirms that the Toll and Imd humoral modules have a defined roles against broad classes of pathogens, Imd for Gram-negative bacteria and DAP-type containing Gram-positive bacteria, and Toll for fungi and Gram-positive bacteria. Use of AMP and Bomanin mutants revealed that this can be largely explained by the effectors they control. In contrast, the contributions of Phagocytosis and Melanization appear to be critical to more specific and diverse sets of pathogens. We speculate that Phagocytosis and/or Melanization have critical roles in defense against bacteria that resist host defense peptides (Hanson et al., 2019), or can hide from them (Touré et al., 2023). The melanization reaction is a source of ROS that is a potent defense against pathogens such as *S. aureus* resistant to Toll and Imd defense (Dudzic et al., 2019; Ford and King, 2021; Needham et al., 2004; Ramond et al., 2021). These two modules play important roles in immune-related processes such as encapsulation (Melanization), the uptake of bacteria escaping from the gut, or tissue homeostasis (Phagocytosis) (Braun et al., 1998; C. Melcarne et al., 2019; Nehme et al., 2007) that were not assessed here. Collectively, our study validates, with minor discrepancies, many studies that have assessed the individual contributions of these modules (e.g. Apidianakis et al., 2005; Binggeli et al., 2014; Lamiabile et al., 2016; Lemaitre et al., 1996, 1995; Nehme et al., 2011). Our mutant lines can now be used to analyze the contribution of these immune modules in resistance to other pathogens, notably wasps, nematodes, microsporidia and protozoans, or in other contexts such as mating and local infection.

While defense against some pathogens is largely mediated by single immune modules, other pathogens are handled by multiple modules. We hypothesize that pathogens that have intermediate virulence levels (kill only a fraction of wild-type flies), may better reveal the roles of multiple modules. Indeed, stochasticity in survival analyses partly stems from the arms-race occurring between the pathogen and host immunity, as shown for *P. rettgeri* (Duneau et al., 2017). In this scenario, any small change in immune competence may tip the outcome of the arms-race toward lethality or survival. Here, it is notable that multiple modules contribute to survival in *P. rettgeri* infection. Previous studies have revealed a major role of the Imd pathway AMP Diptericin against this bacterium (Hanson et al., 2023, 2019; Unckless et al., 2016). However, Duneau et al. (Duneau et al., 2017) showed that survival patterns to *P. rettgeri* bifurcate into two outcomes based on time taken to fully activate systemic defenses, and showed a role for a Toll-PO SP cascade regulating serine proteases in defense against this microbe (Duneau

et al., 2017 [↗](#)). We may speculate that Melanization, although not as great a contributor as Dipterecin, may be a factor that tips the balance of this precarious arms race towards host lethality. The observation that $\Delta Phag$ flies expressed less *DptA* (**Figure 1** [↗](#)) can explain their susceptibility to *P. rettgeri*. Future studies should investigate how multiple modules can contribute to host survival according to additive or Achilles dynamics – the concept that microbes have specific weaknesses available for host effectors to target (Hanson, 2024 [↗](#)). It will also be important to consider the distinct roles of resistance, tolerance, and resilience in host defense (Howick and Lazzaro, 2017 [↗](#); Wukitch et al., 2023 [↗](#)).

We observed a good correlation between survival analysis and pathogen growth in single module flies for *P. rettgeri*, *S. aureus*, and *C. albicans*. This indicates that these immune modules mostly contribute to resistance mechanisms that suppress pathogen growth. Our study did not reveal key early contributions of phagocytosis or melanization in pathogen growth control, despite quasi-immediate activation of these modules; higher *S. aureus* growth at 2 hours in $\Delta Phag$ flies was paralleled by ΔIMD flies, and we observed reduced Imd activation in $\Delta Phag$ flies. Melanization, while critical to surviving *S. aureus* infection, impacts bacterial growth beginning only at the 12-hour time point. This indicates that the microbicidal activity associated with Melanization in vivo acts more slowly than the blackening reaction observed in bled hemolymph. Interestingly, $\Delta Phag$ and also ΔMel flies suppress *C. albicans* yeast growth, but some individuals ultimately succumb to infection with lower PLUD levels. We confirmed $\Delta Phag$ PLUD results using both an isogenic and a second wild-type genetic background, suggesting that this lower PLUD is genuine. Susceptibility to fungal infection independent of fungal proliferation has also been reported using an *A. fumigatus* septic infection model, and relies on the protection offered by Bomanins from pathogen-secreted toxins (Xu et al., 2023 [↗](#)). It is tempting to speculate, based on the modules involved, that the loss of tolerance to *C. albicans* we observe is related to wound repair or accumulating damage, similar to reports of renal failure in beetles (Khan et al., 2017 [↗](#); Li et al., 2020 [↗](#)), or flies with autotoxic trachea degradation upon stress pathway disruption (Rommelaere et al., 2024 [↗](#)). Thus, this tolerance effect could rely on pathogen-mediated or autotoxic damage, which may be elucidated in a future study.

Here we have provided a single and double mutant analysis of *Drosophila* immune module functions. Our study provides several insights showing which modules are most important to survive infection by defined pathogens. However, we also highlight collective contribution of modules to defense, even when one module is of outsized importance. We extend our comprehension of innate immune responses by revealing higher complexity in defense, implicating multiple host immune modules in survival to various germs, including some with more cryptic contributions. As illustrated by our previous characterizations of AMP function (Carboni et al., 2022 [↗](#); Hanson et al., 2019 [↗](#)), the melanization response (Dudzic et al., 2015 [↗](#)), and stress-induced Turandot proteins (Rommelaere et al., 2024 [↗](#)), a combinatorial mutation approach to deciphering immune function can be extended even to the broad level of whole immune modules. Together with these previous studies and others, our study provides an immunity toolkit, including flies simultaneously deficient for four immune modules, which can be used to better characterize the contributions of immune processes to host defense. Our tools can also be of use to study various questions at the cutting edge of aging, memory, neurodegeneration, cancer, and more, where immune genes are repeatedly implicated. We hope that this set of lines will be useful to the community to better characterize the *Drosophila* host defense.

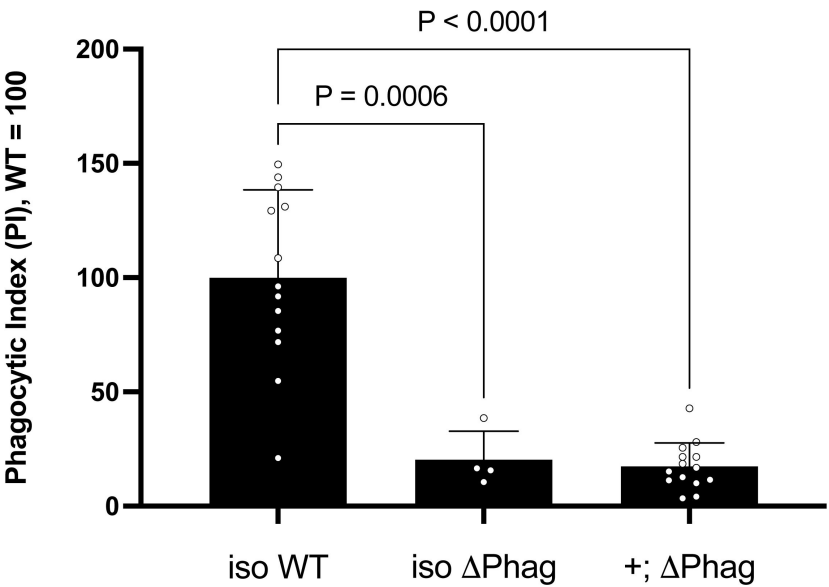
Table S1

Microbe characteristics and growth conditions.

Type	Kingdom	Phylum	Class	Order	Family	Nucleic acid / Membrane	Species	Strain, Source, DOI of previous study	Inf. mode	Infection temperature	Used concentration	Growth temperature	Medium of culture
Viruses	Bamfordvirae	Nucleocytoviricota	Megaviricetes	Pimascovirales	Iridoviridae	dsDNA+	<i>Invertebrate iridescent 5 virus</i>	gift from Ronald van Rij	intra-thoracic	25°C	31 700 TCID50	provided	provided
	Orthomavirae	nd	nd	nd	Bimaviridae	ssRNA+	<i>Drosophila X virus</i>	gift from Ronald van Rij	intra-thoracic	25°C	431 000 TCID50	provided	provided
	Orthomavirae	Plasiviricota	Pisoniviricetes	Picomavirales	Dicistroviridae	ssRNA+	<i>Drosophila C virus</i>	gift from Carla Saleh	intra-thoracic	25°C	2 000 TCID50	26°C	S2 cells in Schneider
	Orthomavirae	Kitrinoviricota	Magsaviricetes	Nodamuvirales	Nodaviridae	dsRNA+	<i>Flock House virus</i>	gift from Carla Saleh	intra-thoracic	25°C	250 000 TCID50	26°C	S2 cells in Schneider
	Orthomavirae	Kitrinoviricota	Alsuviricetes	Martellivirales	Togaviridae	ssRNA+	<i>Sindbis virus</i>	gift from Ronald van Rij	intra-thoracic	25°C	50 000 TCID50	provided	provided
Fungi	Fungi	Ascomycota	Sacharomycete	Sacharomycetales	Sacharomycetaceae	Beta-glucans	<i>Candida albicans</i>	ATCC 2001	septic	29°C	OD ⁶⁰⁰ = 200	37°C	YPG
		Ascomycota	Eurotiomycete	Eurotiales	Typhochomaceae	Beta-glucans	<i>Aspergillus fumigatus</i>	doi: 10.7554/eLife.44341	natural	29°C	rolled on plate	37°C	Malt-agar
		Ascomycota	Sordariomycete	Corcyptitaceae	Ophiocordycipitaceae	Beta-glucans	<i>Beauveria bassiana</i>	R444	septic	29°C	OD ⁶⁰⁰ = 5	commercial	commercial
		Ascomycota	Sordariomycete	Corcyptitaceae	Ophiocordycipitaceae	Beta-glucans	<i>Beauveria bassiana</i>	R444	natural	29°C	30mg	commercial	commercial
Gram positive bacteria	Bacteria	Actinomycetota	Actinomycetia	Mycobacteriales	Mycobacteriaceae	mAGP-type	<i>Mycobacterium marinum</i>	121BR WT	intra-thoracic	25°C	OD ⁶⁰⁰ = 1	37°C	LB
		Actinomycetota	Actinomycetia	Mycobacteriales	Corynebacteriaceae	Lys-type	<i>Corynebacterium diphtheriae</i>	DSM 44123	septic	25°C	OD ⁶⁰⁰ = 100	37°C	LB
		Actinomycetota	Actinomycetia	Micrococcales	Micrococcaceae	Lys-type	<i>Micrococcus luteus</i>	doi: 10.7554/eLife.44341	septic	25°C	OD ⁶⁰⁰ = 200	29°C	LB
		Bacillota	Bacilli	Lactobacillales	Streptococcaceae	Lys-type	<i>Streptococcus pneumoniae</i>	D39V WT	septic	25°C	OD ⁶⁰⁰ = 0.3	37°C	LB
		Bacillota	Bacilli	Lactobacillales	Enterococcaceae	Lys-type	<i>Enterococcus faecalis</i>	254	septic	25°C	OD ⁶⁰⁰ = 5	37°C	LB
		Bacillota	Bacilli	Caryophanae	Staphylococcaceae	Lys-type	<i>Staphylococcus aureus</i>	doi: 10.7554/eLife.44341	septic	25°C	OD ⁶⁰⁰ = 0.5	37°C	BHI
		Bacillota	Bacilli	Bacilliales	Listeriaceae	DAP-type	<i>Listeria monocytogenes</i>	BUG 1600	septic	25°C	OD ⁶⁰⁰ = 0.5	37°C	LB
		Bacillota	Bacilli	Bacilliales	Bacillaceae	DAP-type	<i>Bacillus subtilis</i>	168	septic	25°C	OD ⁶⁰⁰ = 5	37°C	BHI
		Pseudomonadota	Gammaproteobact	Pseudomonadales	Pseudomonadaceae	DAP-type	<i>Pseudomonas aeruginosa</i>	PAO1	septic	25°C	OD ⁶⁰⁰ = 0.01	37°C	LB
		Pseudomonadota	Gammaproteobact	Vibrionales	Vibrionaceae	DAP-type	<i>Vibrio parahaemolyticus</i>		septic	25°C	OD ⁶⁰⁰ = 10	37°C	LB
Gram negative bacteria	Bacteria	Pseudomonadota	Gammaproteobact	Enterobacteriales	Morgenellaceae	DAP-type	<i>Providencia rettgeri</i>	Dmel	septic	25°C	OD ⁶⁰⁰ = 0.1	37°C	LB
		Pseudomonadota	Gammaproteobact	Enterobacteriales	Morgenellaceae	DAP-type	<i>Providencia burhodograna</i>	B	septic	25°C	OD ⁶⁰⁰ = 10	37°C	LB
		Pseudomonadota	Gammaproteobact	Enterobacteriales	Enterobacteriaceae	DAP-type	<i>Pectobacterium carotovorum</i>	Ecc15	septic	25°C	OD ⁶⁰⁰ = 10	37°C	LB
		Pseudomonadota	Gammaproteobact	Enterobacteriales	Enterobacteriaceae	DAP-type	<i>Klebsiella pneumoniae</i>		septic	25°C	OD ⁶⁰⁰ = 1.0	37°C	LB
		Pseudomonadota	Gammaproteobact	Enterobacteriales	Enterobacteriaceae	DAP-type	<i>Enterobacter cloacae</i>	B12	septic	25°C	OD ⁶⁰⁰ = 10	37°C	LB
		Pseudomonadota	Gammaproteobact	Enterobacteriales	Enterobacteriaceae	DAP-type	<i>Escherichia coli</i>	1106	septic	25°C	OD ⁶⁰⁰ = 200	37°C	LB
		Pseudomonadota	Gammaproteobact	Enterobacteriales	Enterobacteriaceae	DAP-type	<i>Salmonella enterica ser. typhimurium</i>	doi: 10.7554/eLife.44341	septic	25°C	OD ⁶⁰⁰ = 200	37°C	LB
		Pseudomonadota	Gammaproteobact	Enterobacteriales	Enterobacteriaceae	DAP-type	<i>Salmonella enterica ser. typhimurium</i>	doi: 10.7554/eLife.44341	septic	25°C	OD ⁶⁰⁰ = 200	37°C	LB
		Pseudomonadota	Gammaproteobact	Enterobacteriales	Enterobacteriaceae	DAP-type	<i>Salmonella enterica ser. typhimurium</i>	doi: 10.7554/eLife.44341	septic	25°C	OD ⁶⁰⁰ = 200	37°C	LB
		Pseudomonadota	Gammaproteobact	Enterobacteriales	Enterobacteriaceae	DAP-type	<i>Salmonella enterica ser. typhimurium</i>	doi: 10.7554/eLife.44341	septic	25°C	OD ⁶⁰⁰ = 200	37°C	LB

Figure 1supp1

We used $\Delta Phag$ mutants in the genetic background of Melcarne et al. (2019) (+; $\Delta Phag$) in some experiments, and confirm here that these mutants are equally deficient in phagocytic capacity to $\Delta Phag$ flies in the iso DrosDel genetic background ($\Delta Phag$).



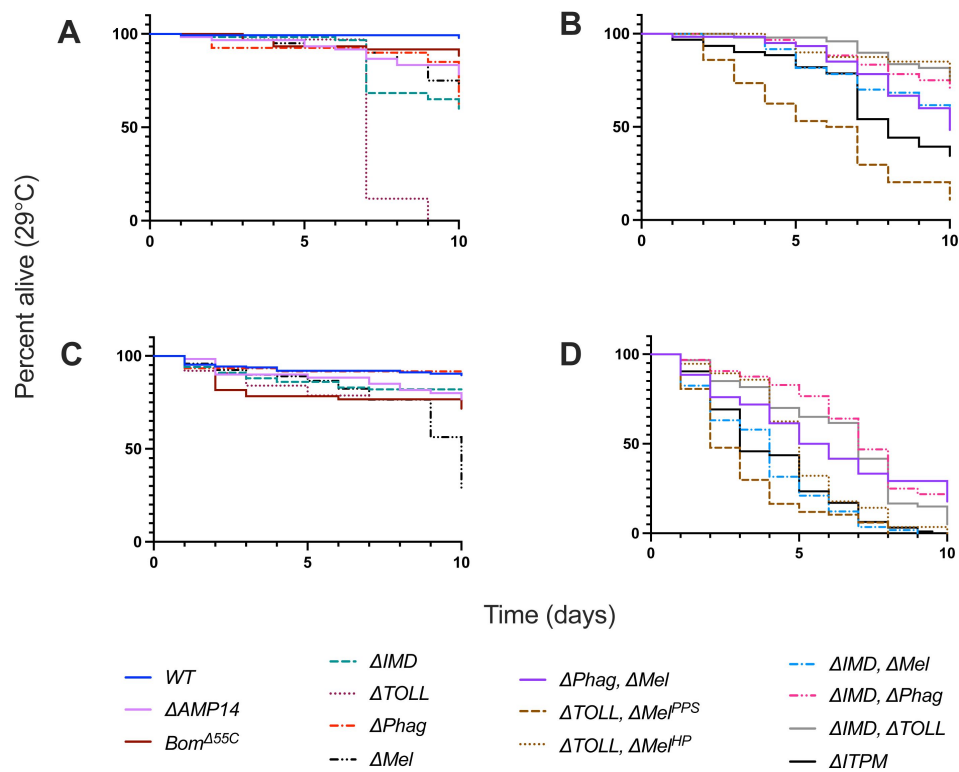


Figure 2supp1

Lifespans of mutants used in this study in unchallenged flies and upon clean injury conditions at 29°C.

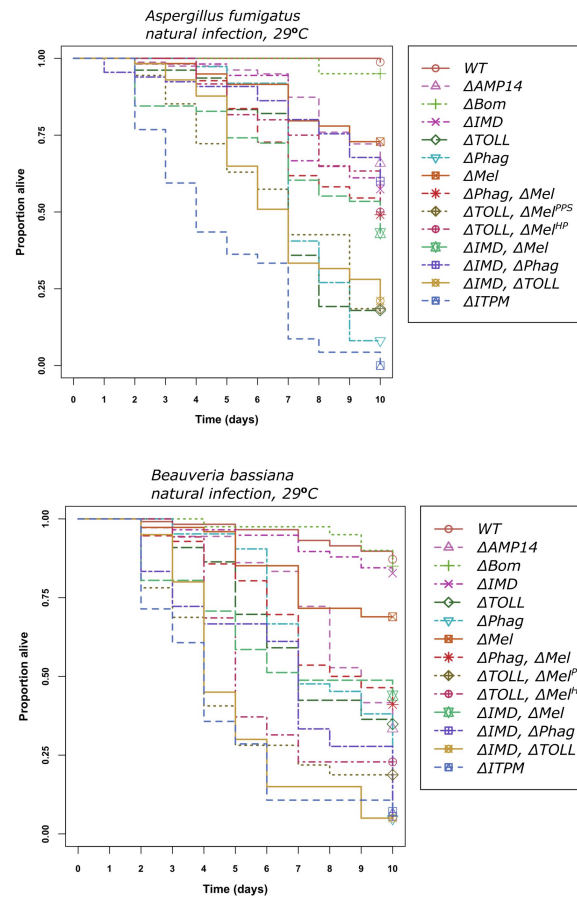
A-B) Single module mutants; **C-D)** compound module mutants.

	Microbe	AMP14	Bom	IMD	TOLL	Phag	Mel
Viruses	DCV	✓		✓	(✓ I, M)		✓
	FHV	✓	✓	✓	✓	✓	✓
	DXV	✓		✓	✓	✓	✓
	IIV-6	✓		✓	(✓ I)		✓
	SINV			(✓ T, P)	(✓ I)	✓	✓
Fungi	<i>A. fumigatus</i> (NI)			(✓ T)	(✓ I, M)		(✓ T)
	<i>B. bassiana</i> (NI)	✓		(✓ P)	(✓ I, M)	(✓ I, M)	(✓ T, P)
	<i>B. bassiana</i> (SI)		✓		✓	✓	
	<i>C. albicans</i> (SI)	✓		(✓ T)	✓		(✓ T)
Gram positive bacteria	<i>M. marinum</i>						
	<i>C. diphtheriae</i>			(✓ T)	(✓ I, M)		✓
	<i>M. luteus</i>			(✓ T)	(✓ I)		
	<i>S. pneumoniae</i>						
	<i>E. faecalis</i>		✓		✓	(✓ M)	(✓ P)
	<i>S. aureus</i>				✓	✓	✓
	<i>L. monocytogenes</i>		✓	✓	✓		
	<i>B. subtilis</i>	✓	✓	✓	✓		(✓ T)
Gram negative bacteria	<i>V. parahemolyticus</i>	✓		✓	✓		
	<i>P. rettgeri</i>	✓		✓	✓	✓	✓
	<i>P. burhodograna</i>	✓		✓	(✓ M)		✓
	<i>Ecc15</i>	✓	✓	✓		✓	
	<i>K. pneumoniae</i>	✓		✓			
	<i>E. cloacae</i>	✓		✓			
	<i>E. coli</i>	✓		✓	(✓ I)	✓	
	<i>S. enterica typhimurium</i>	✓		✓	(✓ I)	(✓ I)	

Figure 3supp1

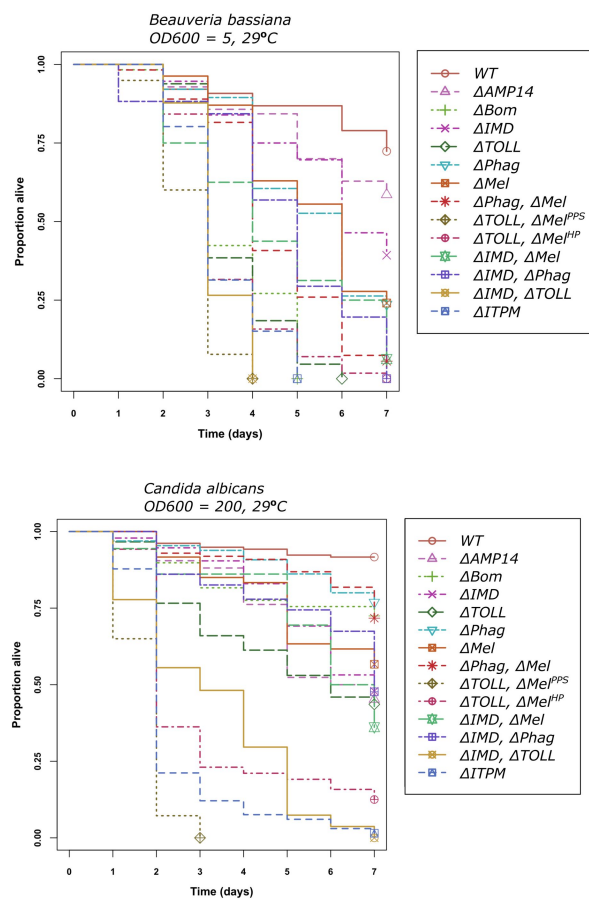
Summary table of susceptibilities per the three conditions outlined in section “Systemic infections and survival.”

✓ indicates that the pathway contributes to defense against a given pathogen. In the event that a single module mutant is not more susceptible to the infection, but contributes to increased mortality in a double mutant line, a ✓ is given and the co-deleted relevant pathways highlighted by their first letter. Black boxes indicate no difference to wild-type or to clean injury. NI, natural infection; SI, septic infection.



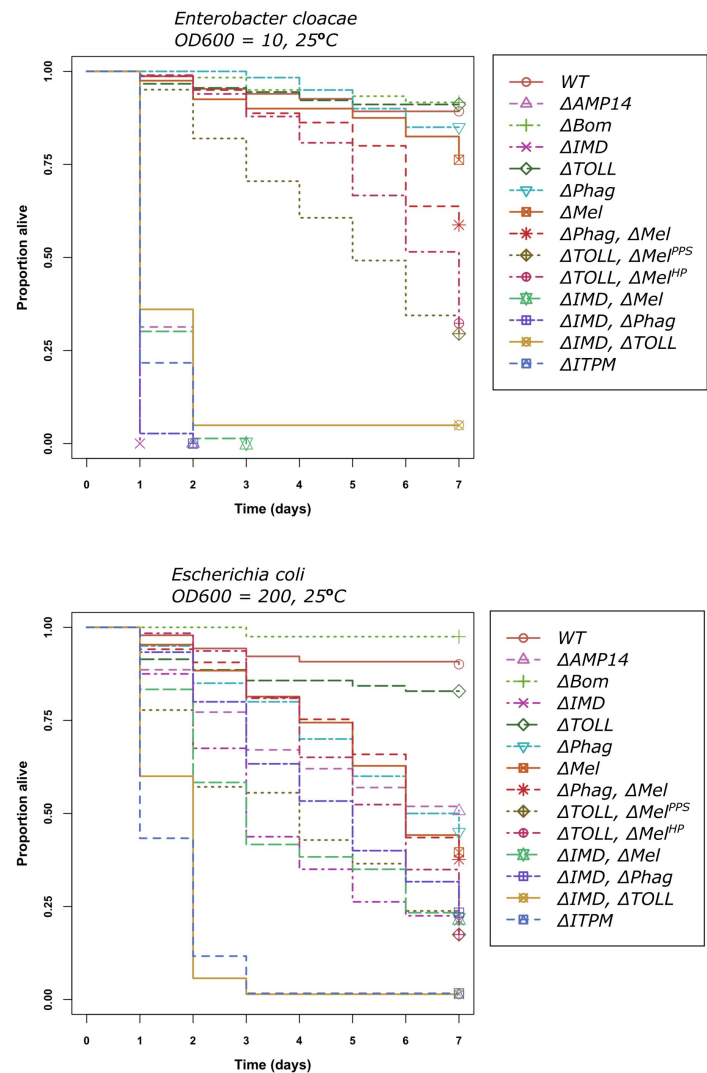
Supplementary file 1

Survival curves for all data summarized in Figure 3 [↗](#).

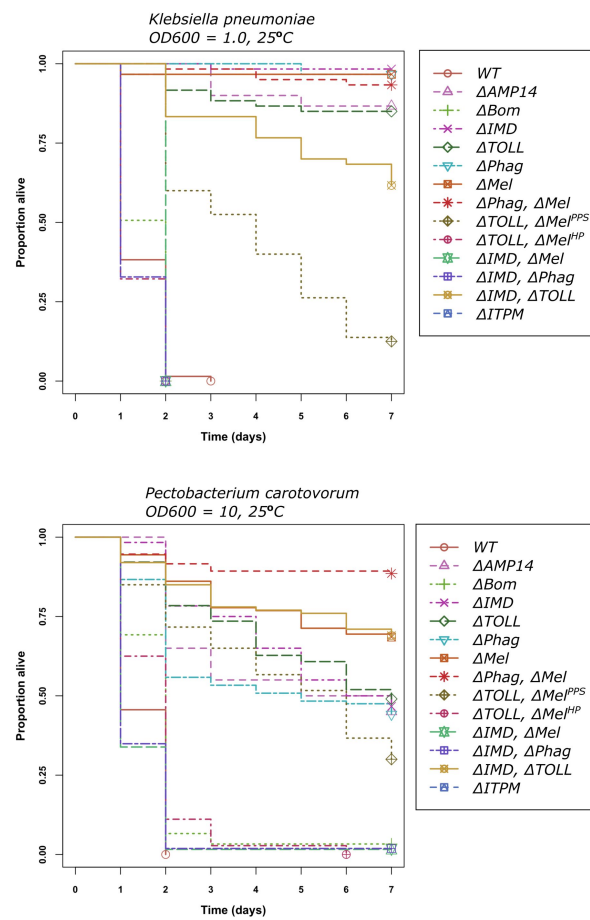


Supplementary file 1 (continued)

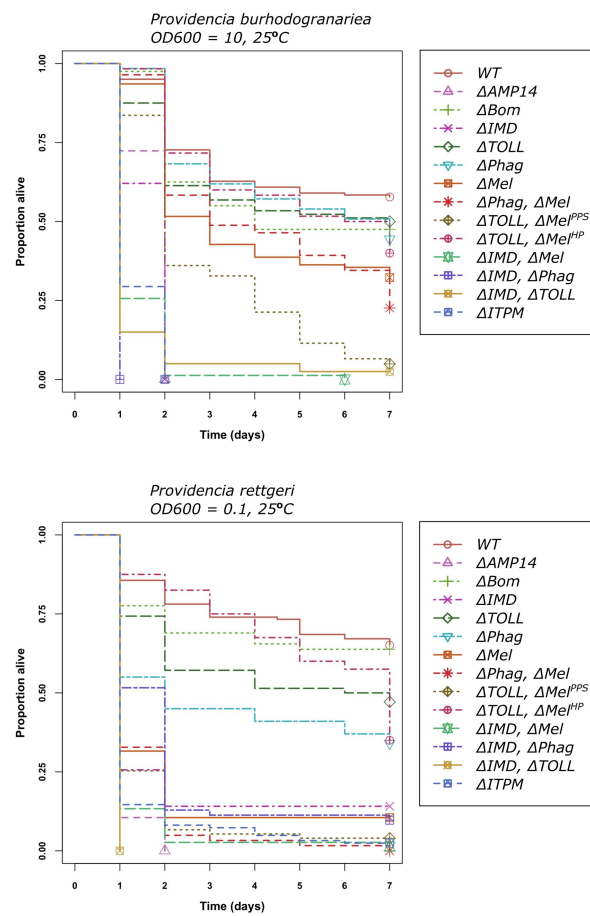
Gram-negative bacteria



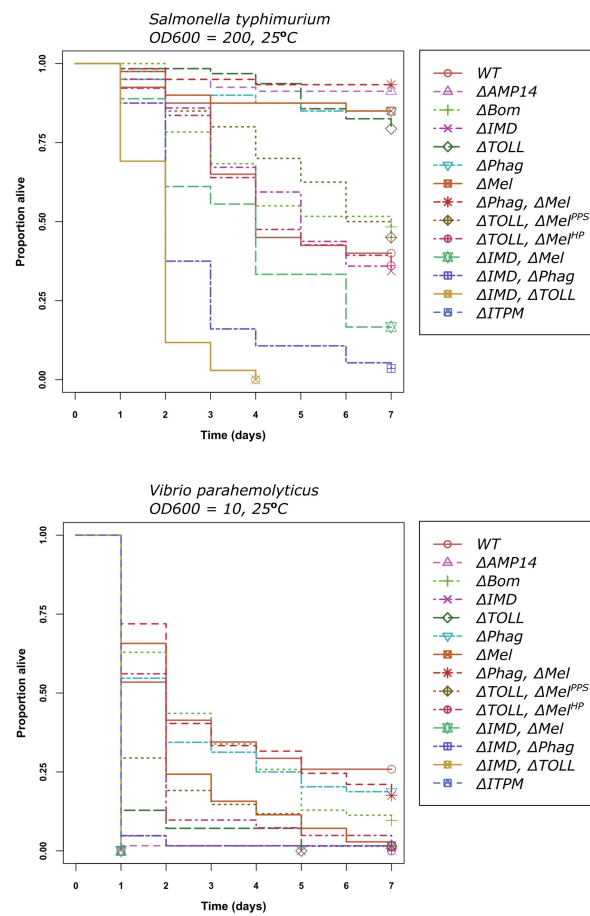
Supplementary file 1 (continued)



Supplementary file 1 (continued)

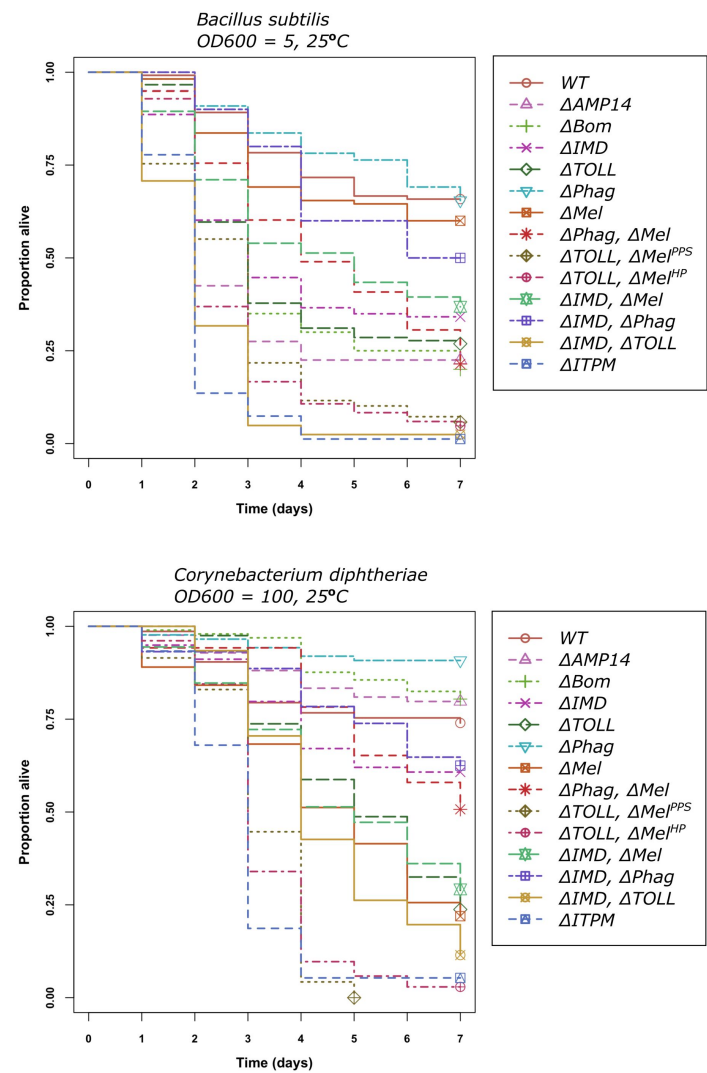


Supplementary file 1 (continued)

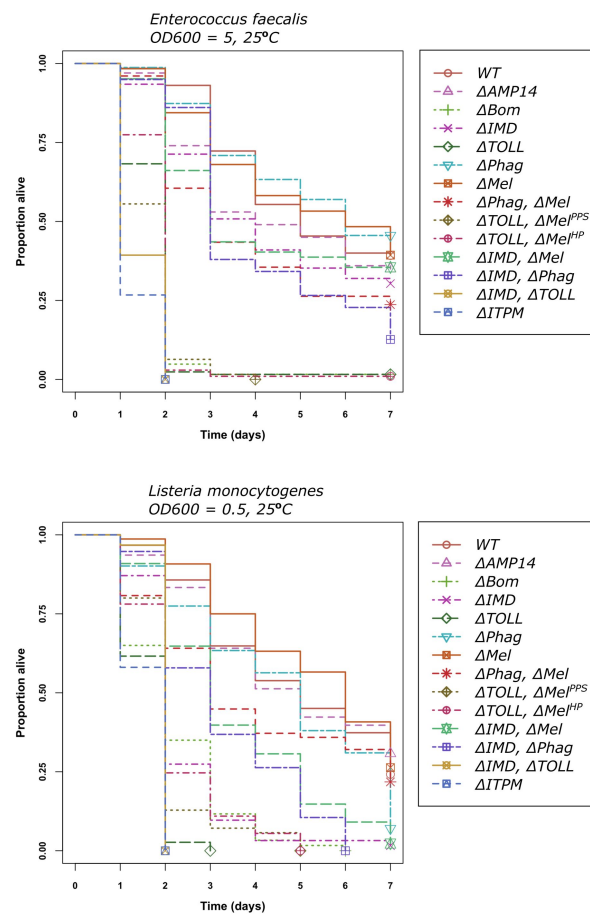


Supplementary file 1 (continued)

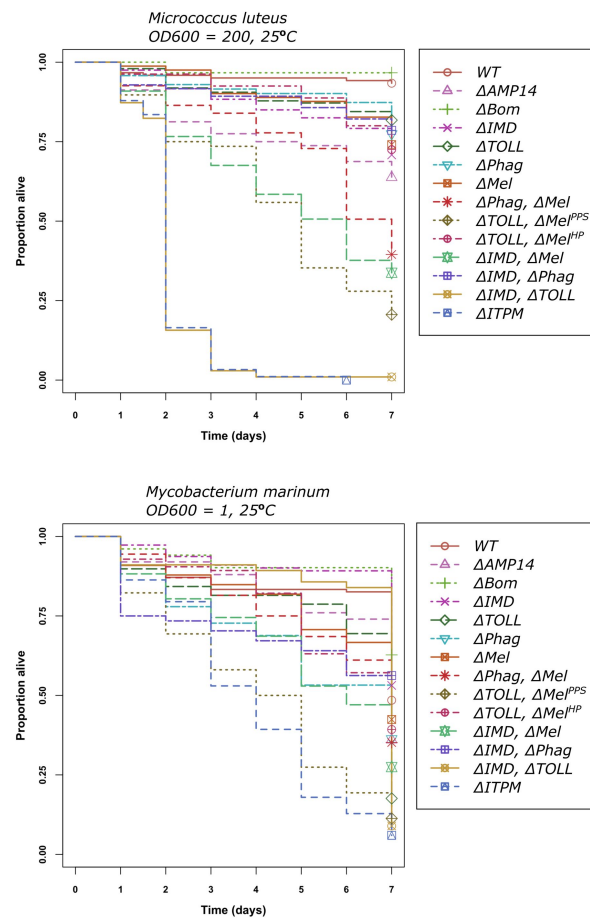
Gram-positive bacteria



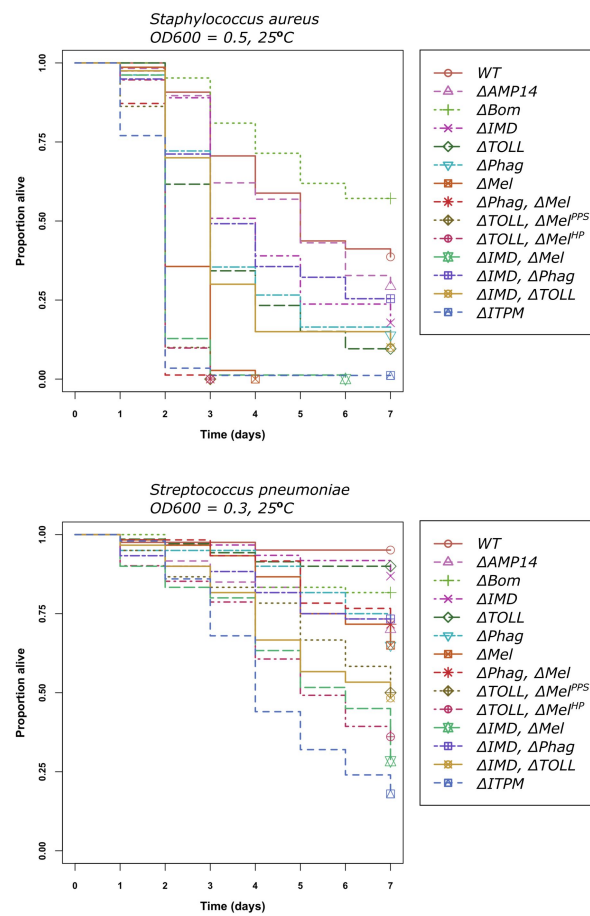
Supplementary file 1 (continued)



Supplementary file 1 (continued)

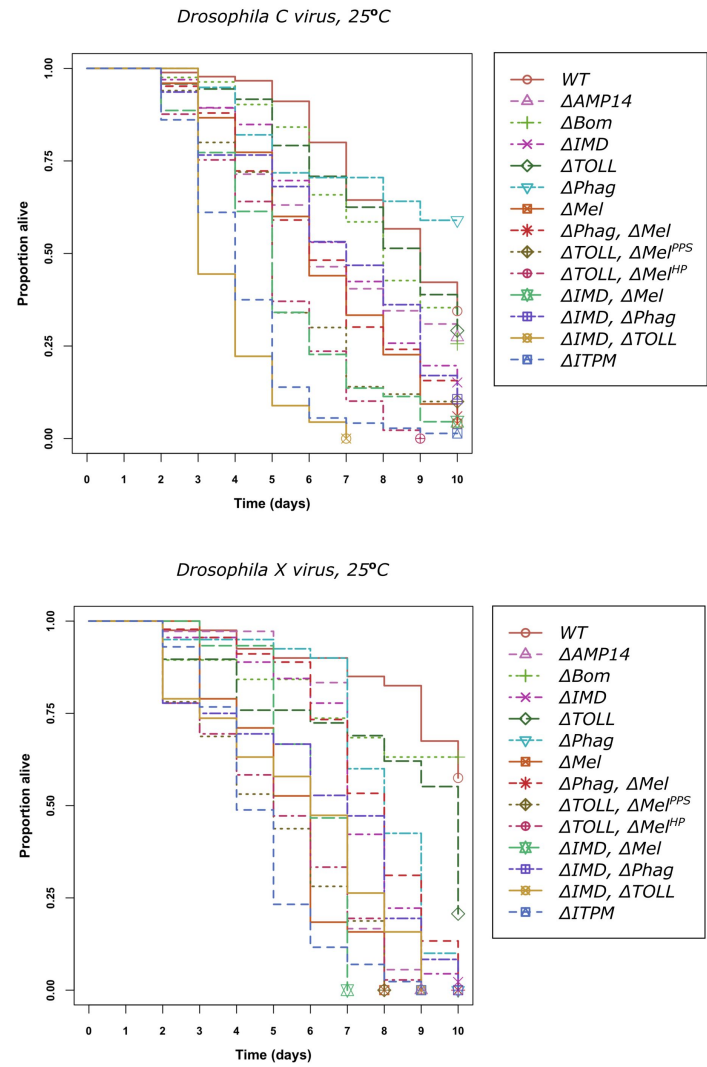


Supplementary file 1 (continued)



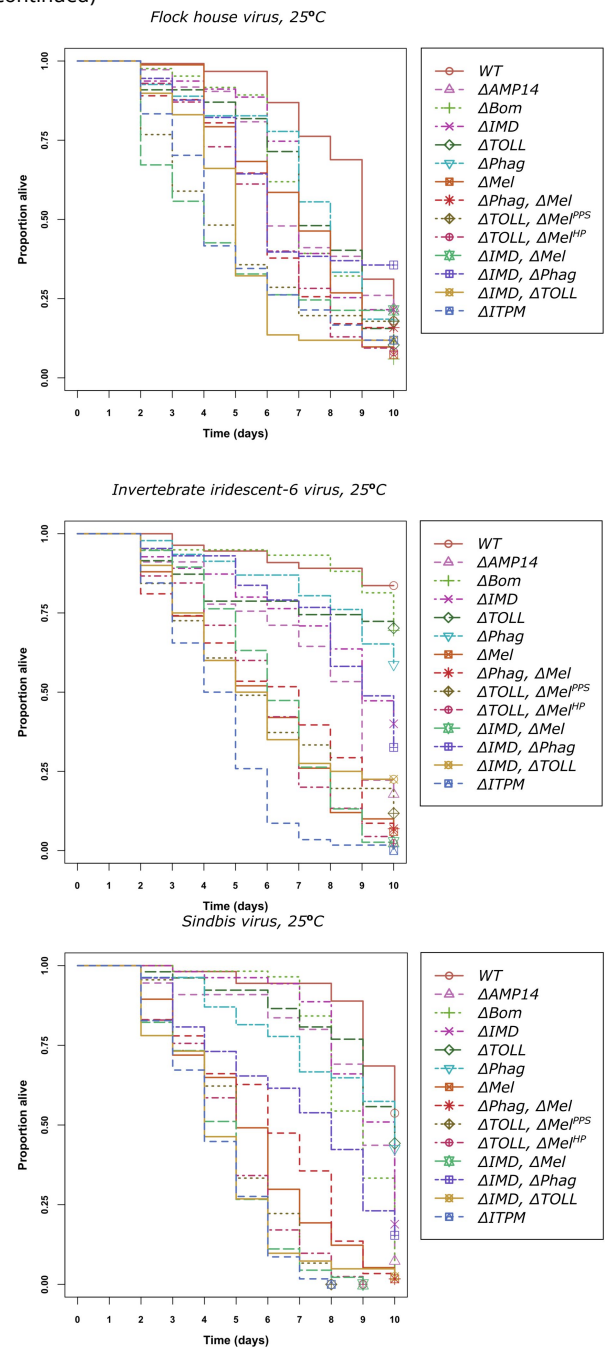
Supplementary file 1 (continued)

Viruses



Supplementary file 1 (continued)

Supplementary file 1 (continued)



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Additional information

Author contributions

Conceptualization: FR, MAH, BL

Data curation: FR, MAH

Funding acquisition: BL, MAH

Investigation: FR, YT, MR, FS

Methodology: FR

Project administration: BL

Validation: YT, MAH

Resources: FR, MAH

Supervision: MAH, BL

Writing – original draft: FR, BL

Writing – review & editing: MAH, BL

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Author information

Faustine Ryckebusch

Global Health Institute, Ecole Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland
ORCID iD: [0000-0002-9626-884X](https://orcid.org/0000-0002-9626-884X)

Yao Tian

Global Health Institute, Ecole Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland
ORCID iD: [0009-0009-6030-5299](https://orcid.org/0009-0009-6030-5299)

Mylene Rapin

Global Health Institute, Ecole Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland

Fanny Schüpfer

Global Health Institute, Ecole Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland

Mark A Hanson*

Global Health Institute, Ecole Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland, Centre for Ecology and Conservation, University of Exeter, Penryn, United Kingdom

ORCID iD: [0000-0002-6125-3672](https://orcid.org/0000-0002-6125-3672)

For correspondence: M.Hanson@exeter.ac.uk

*These authors contributed equally to this work

Bruno Lemaitre*

Global Health Institute, Ecole Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland

ORCID iD: [0000-0001-7970-1667](https://orcid.org/0000-0001-7970-1667)

For correspondence: bruno.lemaitre@epfl.ch

*These authors contributed equally to this work

Editors

Reviewing Editor

Jiwon Shim

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Senior Editor

Satyajit Rath

Indian Institute of Science Education and Research (IISER), Pune, India

Reviewer #1 (Public review):

Summary:

The innate immune system serves as the first line of defense against invading pathogens. Four major immune-specific modules - the Toll pathway, the Imd pathway, melanization, and phagocytosis- play critical roles in orchestrating the immune response. Traditionally, most studies have focused on the function of individual modules in isolation. However, in recent years, it has become increasingly evident that effective immune defense requires intricate interactions among these pathways.

Despite this growing recognition, the precise roles, timing, and interconnections of these immune modules remain poorly understood. Moreover, addressing these questions represents a major scientific undertaking.

Strengths:

In this manuscript, Ryckebusch et al. systematically evaluate both the individual and combined contributions of these four immune modules to host defense against a range of pathogens. Their findings significantly enhance our understanding of the layered architecture of innate immunity.

Weaknesses:

While I have no critical concerns regarding the study, I do have several suggestions to offer that may help further strengthen the manuscript. These include:

(1) Have the authors validated the efficiency of the mutants used in this study? It would be helpful to include supporting data or references confirming that the mutations effectively disrupted the intended immune pathways.

(2) Given the extensive use of double, triple, and quadruple mutants, a more detailed description of the mutant construction process is warranted.

<https://doi.org/10.7554/eLife.107030.1.sa1>

Reviewer #2 (Public review):

Summary:

In this work, the authors take a holistic view of *Drosophila* immunity by selecting four major components of fly immunity often studied separately (Toll signaling, Imd signaling, phagocytosis, and melanization), and studying their combinatory effects on the efficiency of the immune response. They achieve this by using fly lines mutant for one of these components, or modules, as well as for a combination of them, and testing the survival of these flies upon infection with a plethora of pathogens (bacterial, viral, and fungal).

Strengths:

It is clear that this manuscript has required a large amount of hands-on work, considering the number of pathogens, mutations, and timepoints tested. In my opinion, this work is a very welcome addition to the literature on fly immune responses, which obviously do not occur in one type of response at a time, but in parallel, subsequently, and/or are interconnected. I find that the major strength of this work is the overall concept, which is made possible by the mutations designed to target the specific immune function of each module (at least seemingly) without major effects on other functions. I believe that the combinatory mutants will be of use for the fly community and enable further studies of the interplay of these components of immune response in various settings.

To control for the effects arising from the genetic variation other than the intended mutations, the mutants have been backcrossed into a widely used, isogenized *Drosophila* strain called w1118. Therefore, the differences accounted for by the genotype are controlled.

I also appreciate that the authors have investigated the two possible ways of dealing with an infection: tolerance and resistance, and how the modules play into those.

Weaknesses:

While controlling for the background effects is vital, the w1118 background is problematic (an issue not limited to this manuscript) because of the wide effects of the white mutation on several phenotypes (also other than eye color/eyesight). It is a possibility that the mutation influences the functionality of the immune response components, for example, via effects of the faulty tryptophan handling on the metabolism of the animal.

I acknowledge that it is not reasonable to ask for data in different backgrounds better representing a "wild type" fly (however, that is defined is another question), but I think this matter should be brought up and discussed.

The whole study has been conducted on male flies. Immune responses show quite extensive sex-specific variation across a variety of species studied, also in the fly. But the reasons for this variation are not fully understood. Therefore, I suggest that the authors conduct a subset of experiments on female flies to see if the findings apply to both sexes, especially the infection-specificity of the module combinations.

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